Nama : **Nicholas Saputra** **Nim :** **065002300019**		**MODUL 13** **Nama Dosen :** **Ir. Teddy Siswanto, MMSi** **Shabrina Teruri, S.Kom., MTI** **Disusun oleh:** **Faiz Kumara, S.Kom**
Hari dan Tanggal : **Rabu, 21 Mei 2025**	**Praktikum Datawarehouse**	**Nama Asisten Laboratorium :** 1. **Nurafni Revita Wibowo - 065002100013** 2. **Siti Aisah - 065002100036**

DATA MODELING

DAN

PROSES EXTRACT, TRANSFORM, LOAD (ETL)

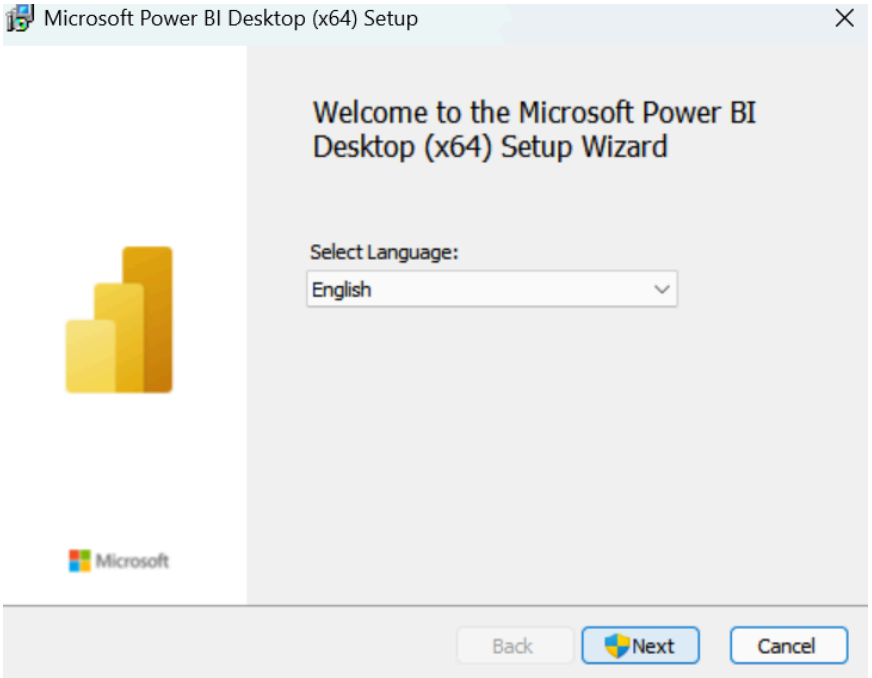
DENGAN POWER BI

ELEMEN KOMPETENSI I

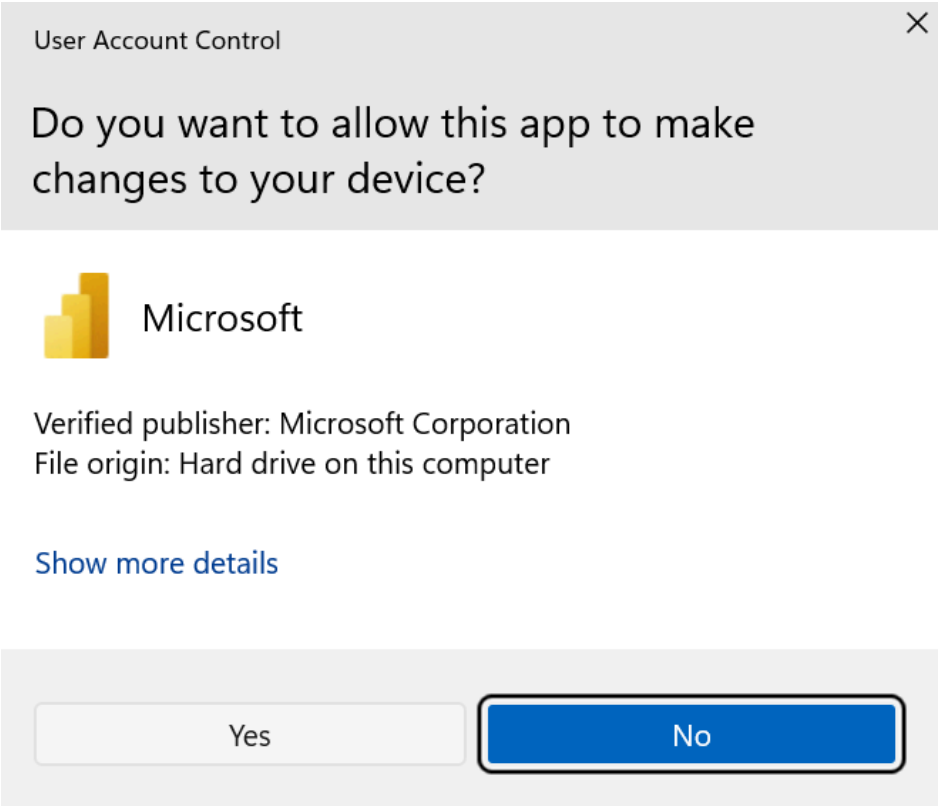
1. Download PowerBI pada link berikut ini :

   PowerBI

   Pilih bahasa sesuai yg diinginkan , lalu klik next pada berikut ini :



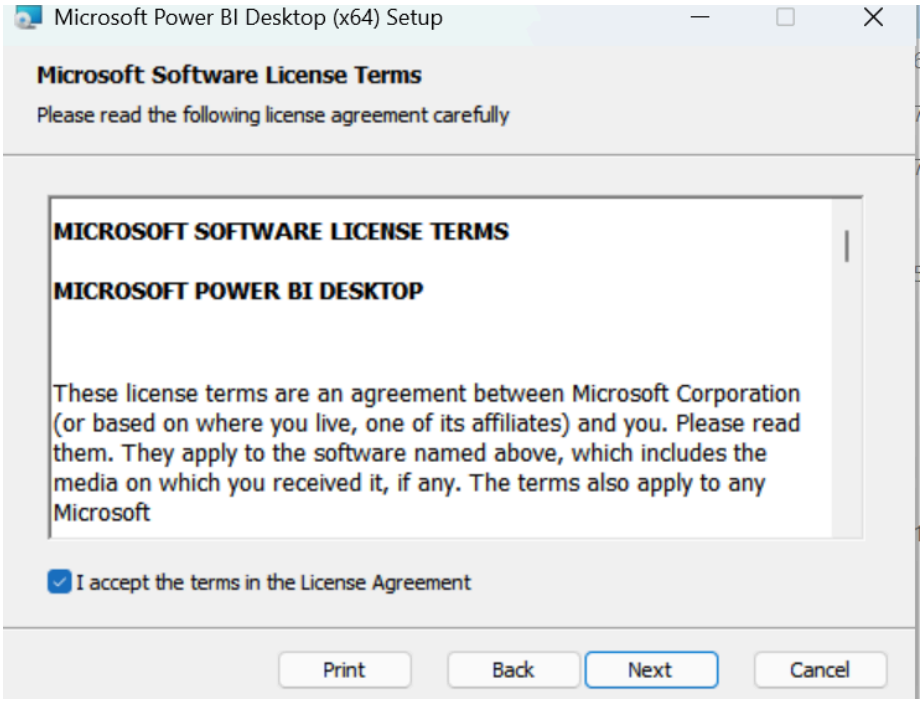
   Lalu klik yes



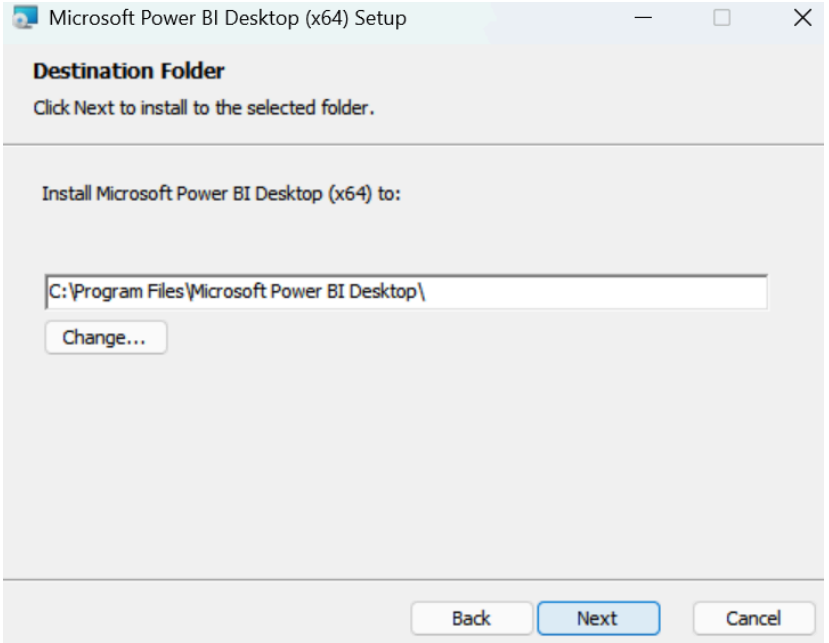
Lalu klik Finish

Kemudian klik next

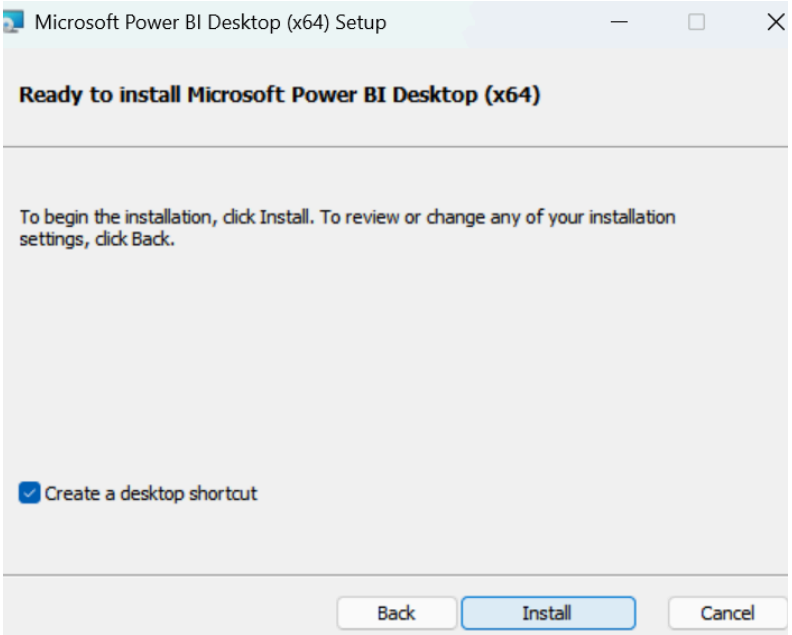
Lalu klik I accept the term, kemudian klik next



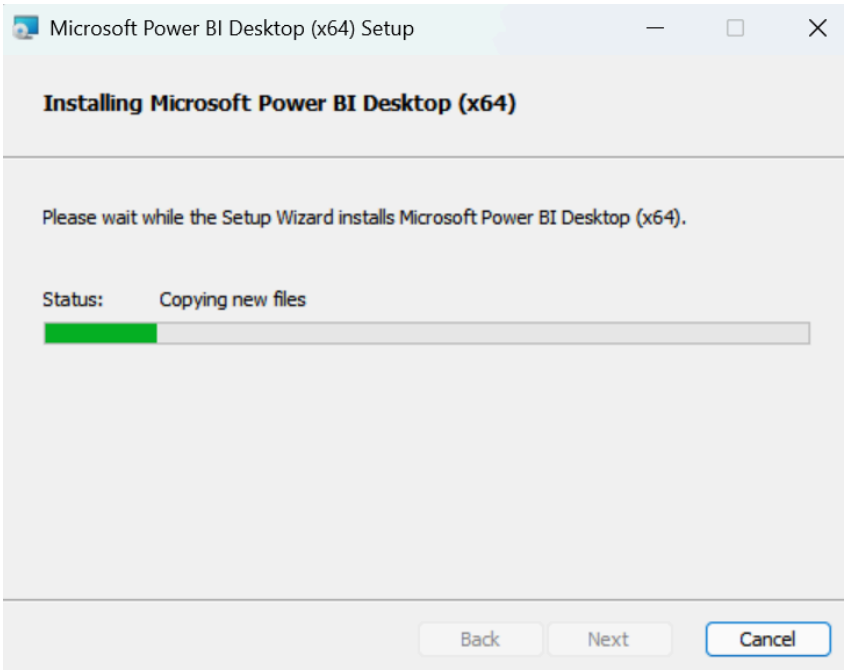
Pilih direktori yang diingkan



Lalu cekliskan, kemudian klik next

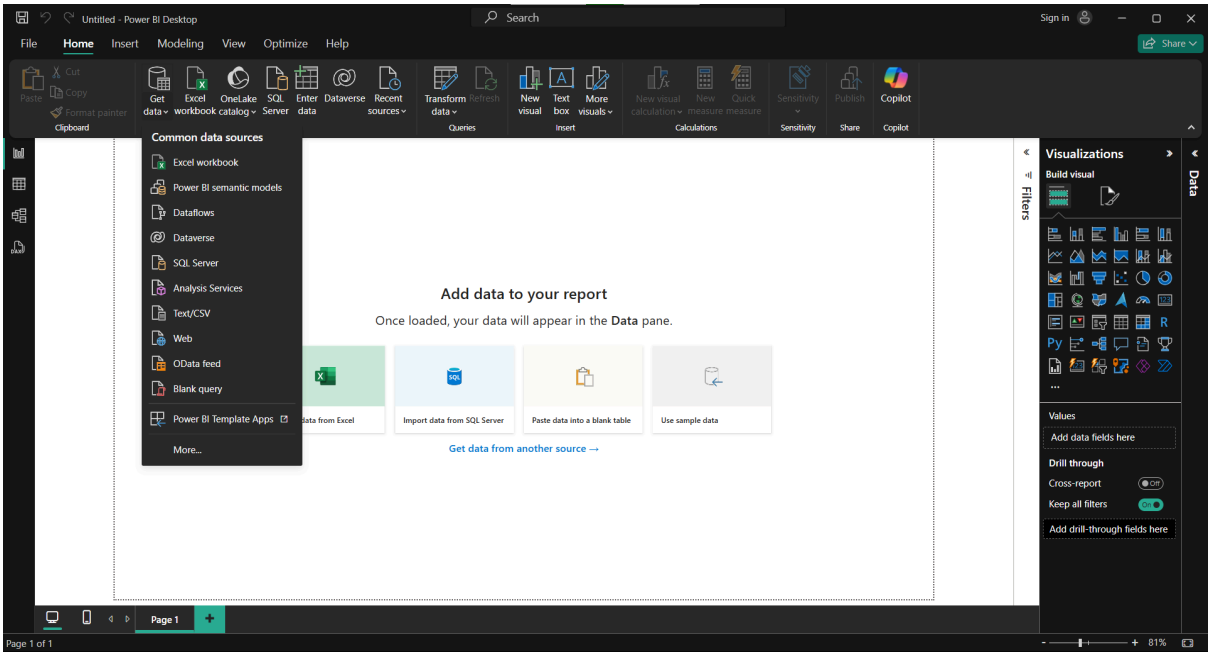


Menunggu sampai instalasi selesai

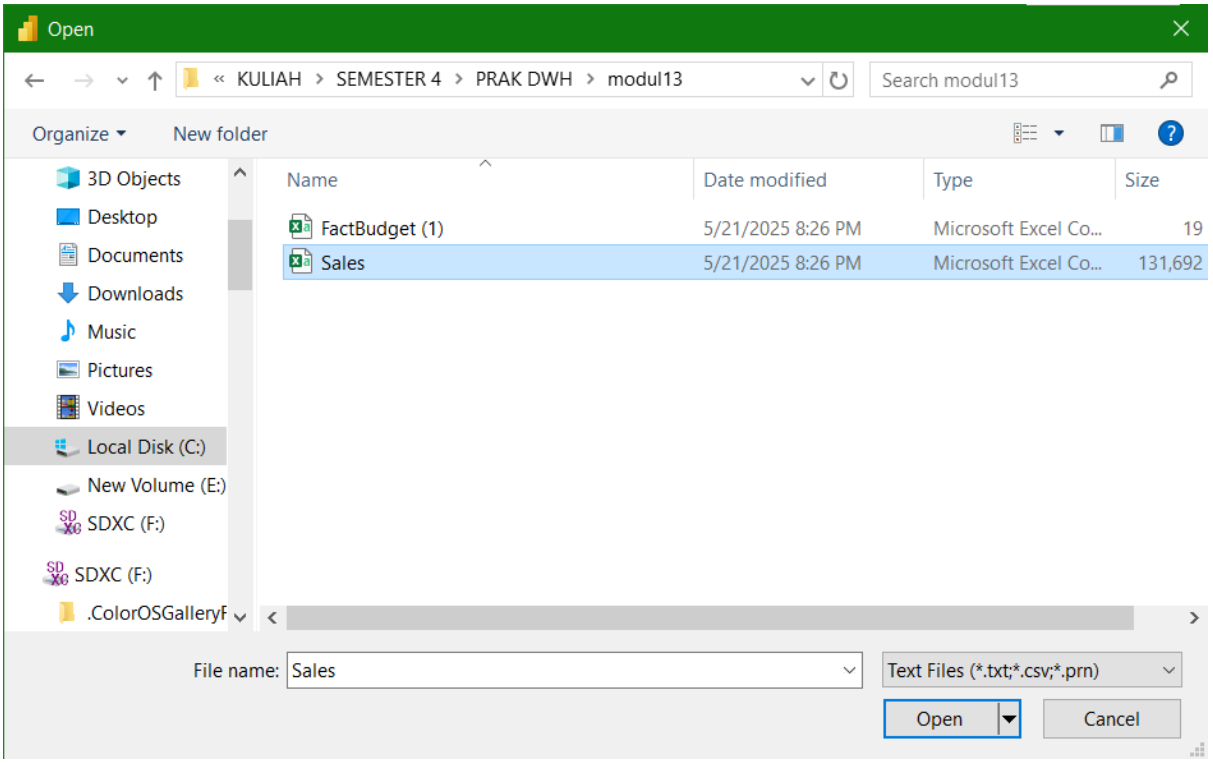


Lalu klik finish

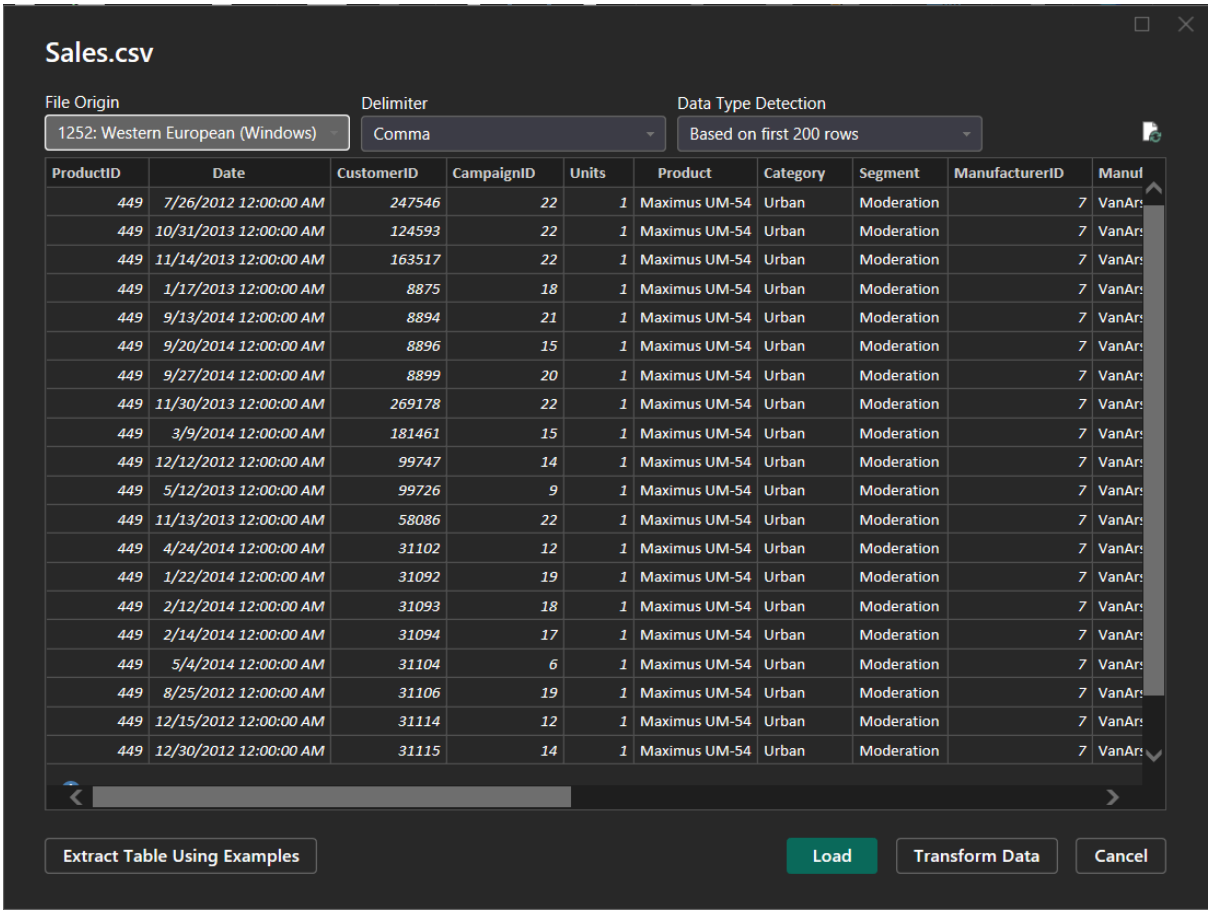
2. Siapkan file berisikan data berbentuk OLTP dan Buka Aplikasi Power BI. Apabila tidak mempunyai aplikasi klik link drive berikut https://drive.google.com/drive/folders/144Sk5olszar_x0Z4302OjHCCKDSL5EPd?usp=sharing
3. Klik breakdown Get Data lalu pilih file Text/CSV



4. Pilih file Sales.csv

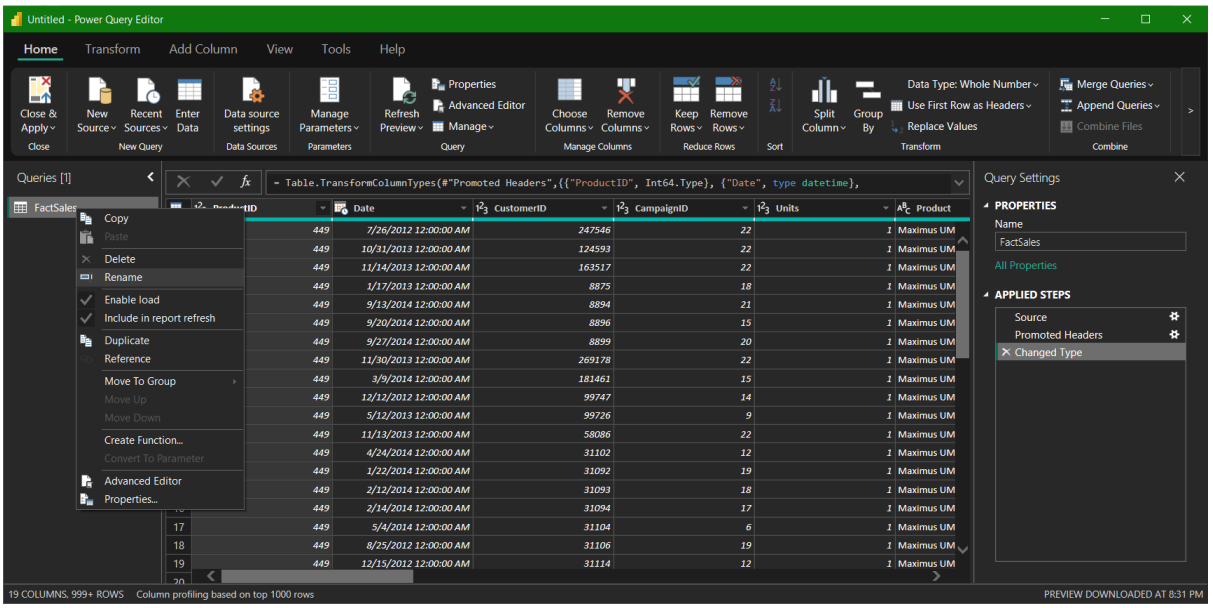


5. Muncul tampilan data yang kita pilih. Karena datanya masih dalam bentuk OLTP kita klik Transform Data untuk mengubahnya menjadi OLAP dengan skema Snowflake

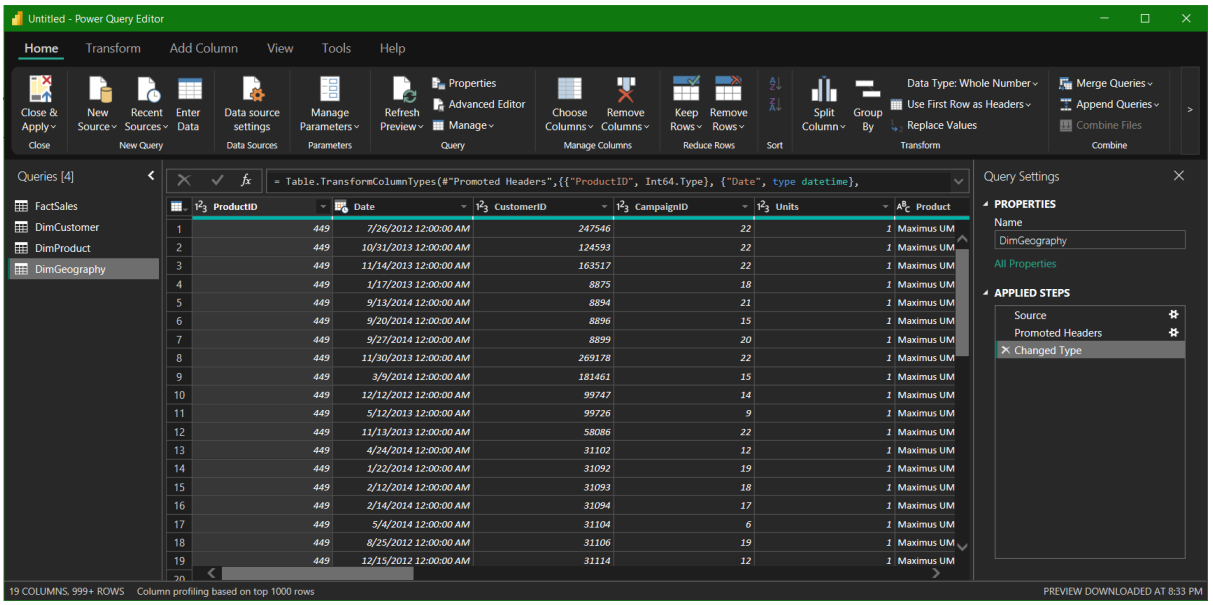


ProductID	Date	CustomerID	CampaignID	Units	Product	Category	Segment	ManufacturerID	Manul
449	7/26/2012 12:00:00 AM	247546	22	1	Maximus UM-54	Urban	Moderation	7	VanArs
449	10/31/2013 12:00:00 AM	124593	22	1	Maximus UM-54	Urban	Moderation	7	VanArs
449	11/14/2013 12:00:00 AM	163517	22	1	Maximus UM-54	Urban	Moderation	7	VanArs
449	1/17/2013 12:00:00 AM	8875	18	1	Maximus UM-54	Urban	Moderation	7	VanArs
449	9/13/2014 12:00:00 AM	8894	21	1	Maximus UM-54	Urban	Moderation	7	VanArs
449	9/20/2014 12:00:00 AM	8896	15	1	Maximus UM-54	Urban	Moderation	7	VanArs
449	9/27/2014 12:00:00 AM	8899	20	1	Maximus UM-54	Urban	Moderation	7	VanArs
449	11/30/2013 12:00:00 AM	269178	22	1	Maximus UM-54	Urban	Moderation	7	VanArs
449	3/9/2014 12:00:00 AM	181461	15	1	Maximus UM-54	Urban	Moderation	7	VanArs
449	12/12/2012 12:00:00 AM	99747	14	1	Maximus UM-54	Urban	Moderation	7	VanArs
449	5/12/2013 12:00:00 AM	99726	9	1	Maximus UM-54	Urban	Moderation	7	VanArs
449	11/13/2013 12:00:00 AM	58086	22	1	Maximus UM-54	Urban	Moderation	7	VanArs
449	4/24/2014 12:00:00 AM	31102	12	1	Maximus UM-54	Urban	Moderation	7	VanArs
449	1/22/2014 12:00:00 AM	31092	19	1	Maximus UM-54	Urban	Moderation	7	VanArs
449	2/12/2014 12:00:00 AM	31093	18	1	Maximus UM-54	Urban	Moderation	7	VanArs
449	2/14/2014 12:00:00 AM	31094	17	1	Maximus UM-54	Urban	Moderation	7	VanArs
449	5/4/2014 12:00:00 AM	31104	6	1	Maximus UM-54	Urban	Moderation	7	VanArs
449	8/25/2012 12:00:00 AM	31106	19	1	Maximus UM-54	Urban	Moderation	7	VanArs
449	12/15/2012 12:00:00 AM	31114	12	1	Maximus UM-54	Urban	Moderation	7	VanArs
449	12/30/2012 12:00:00 AM	31115	14	1	Maximus UM-54	Urban	Moderation	7	VanArs

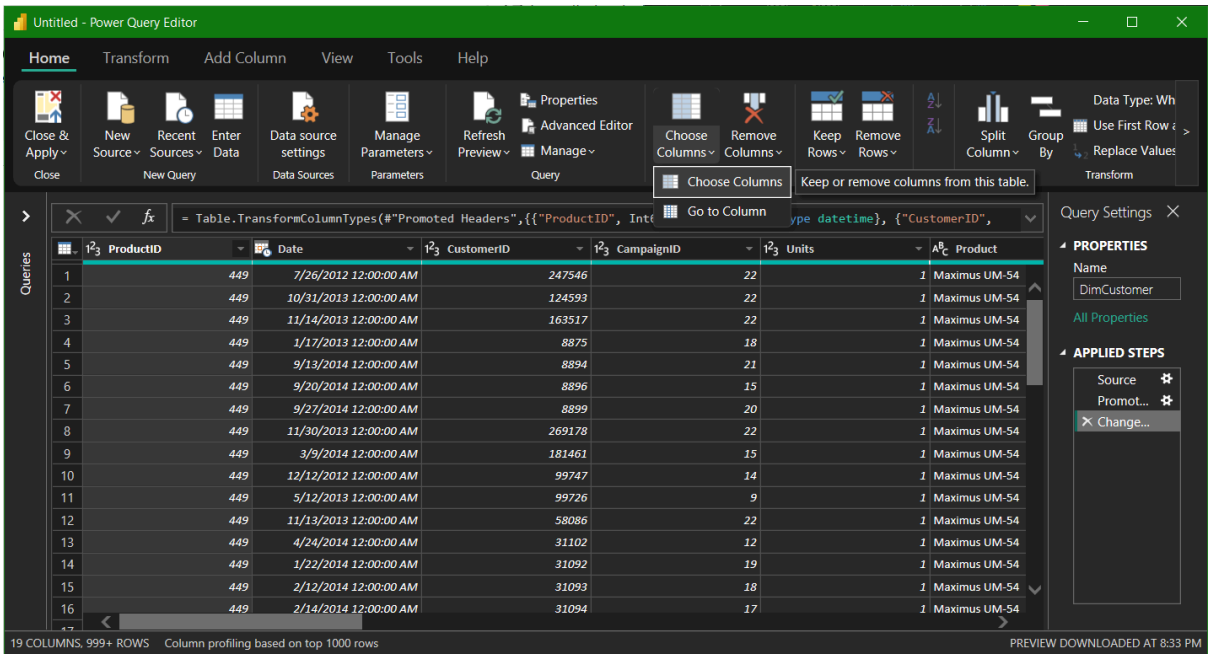
6. Setelah masuk sesi Transform Data, kita perlu rename nama file Sales menjadi FactSales untuk menandakan bahwa ini merupakan tabel fact

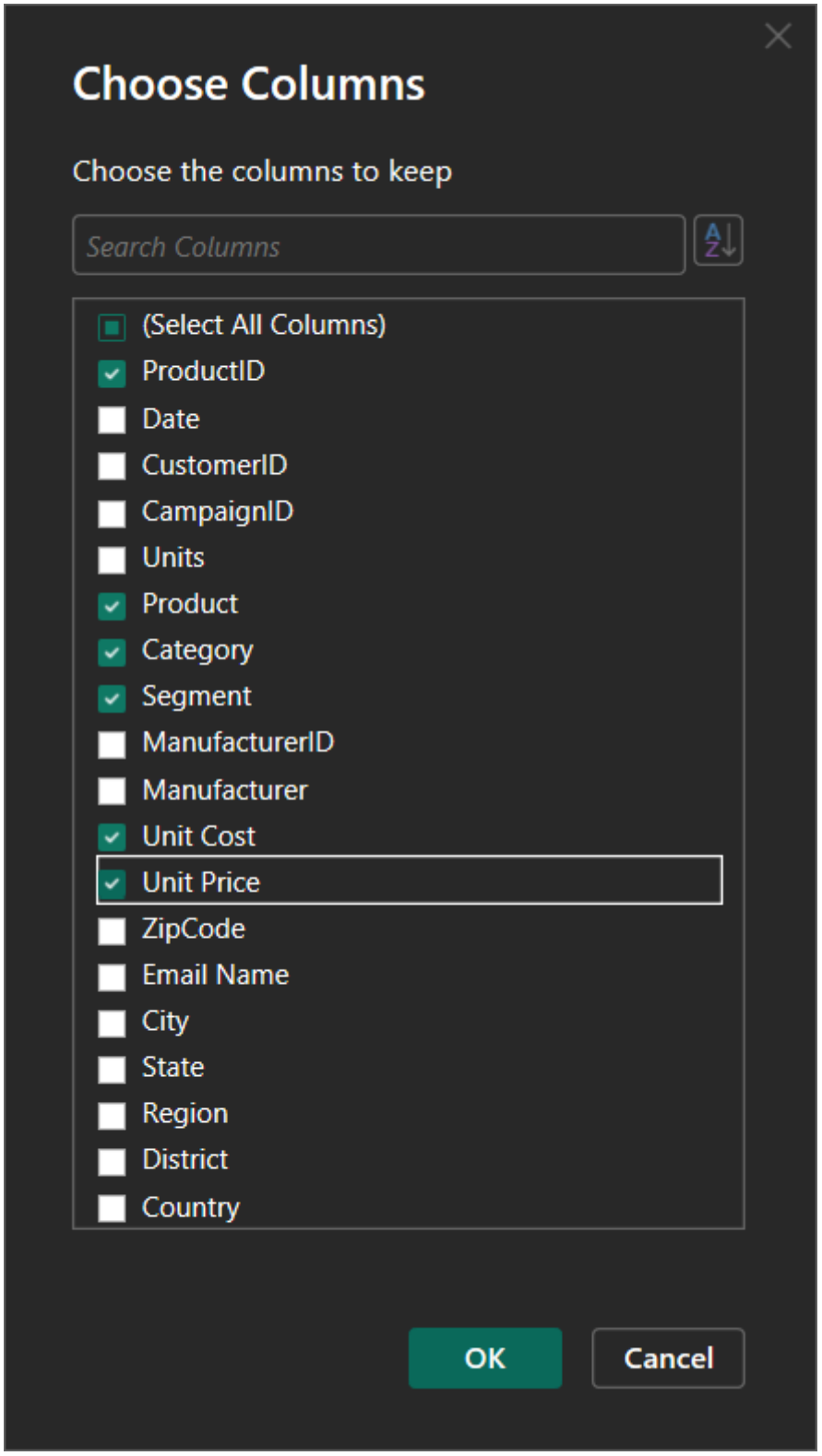


7. Setelah rename, kita lakukan duplikasi tabel sebanyak 3 tabel duplikasi yang masing-masing akan dijadikan tabel dimensional lalu rename tabel duplikasi tadi menjadi DimCustomer, DimProduct, dan DimGeography

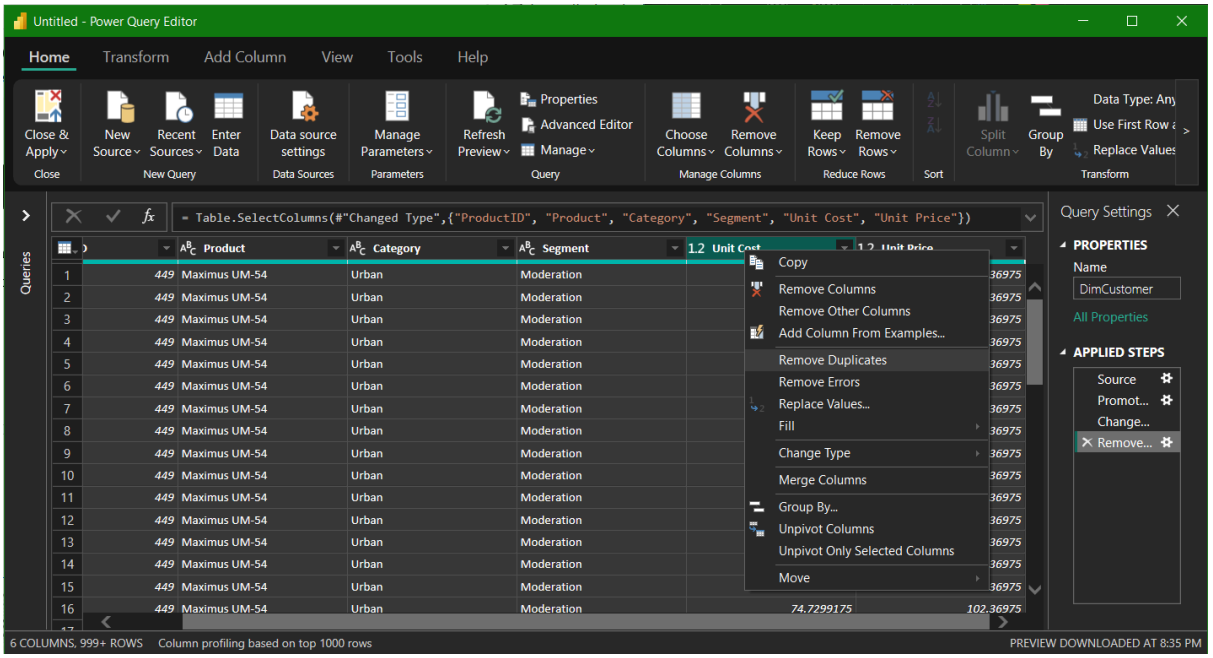


8. Kita beralih ke tabel DimProduct untuk menghilangkan kolom yang tidak dipakai. Klik Choose Column lalu pilih kolom ProductID, Product, Category, Segment, Unit Cost, Unit Price.

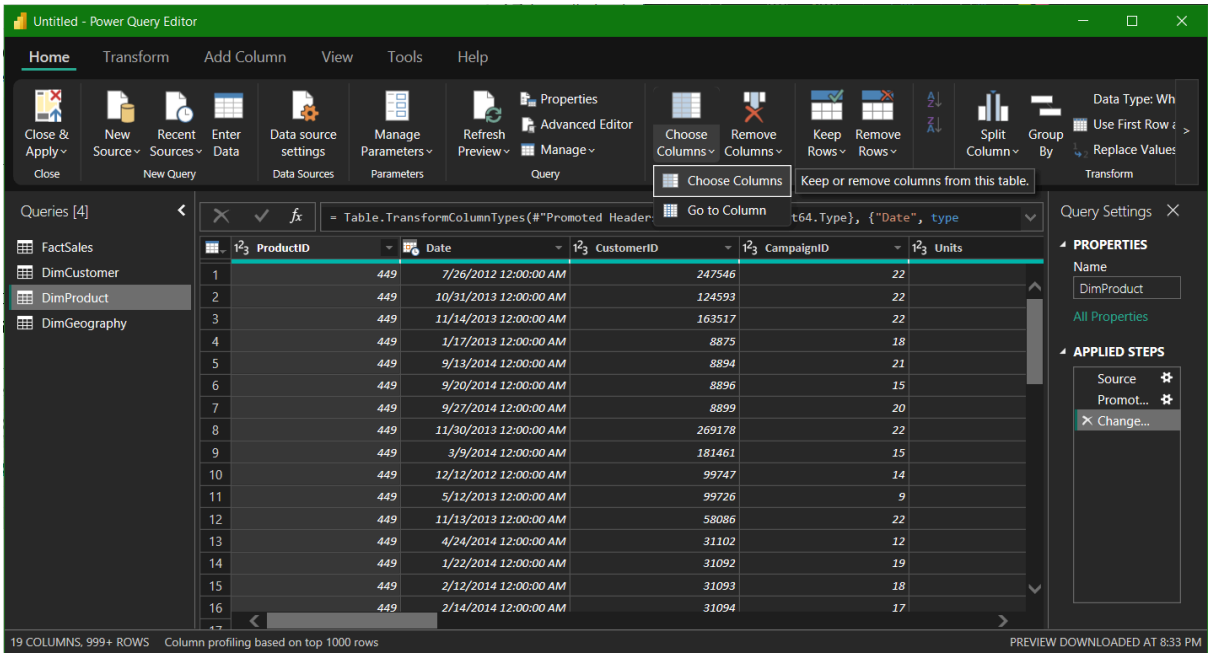


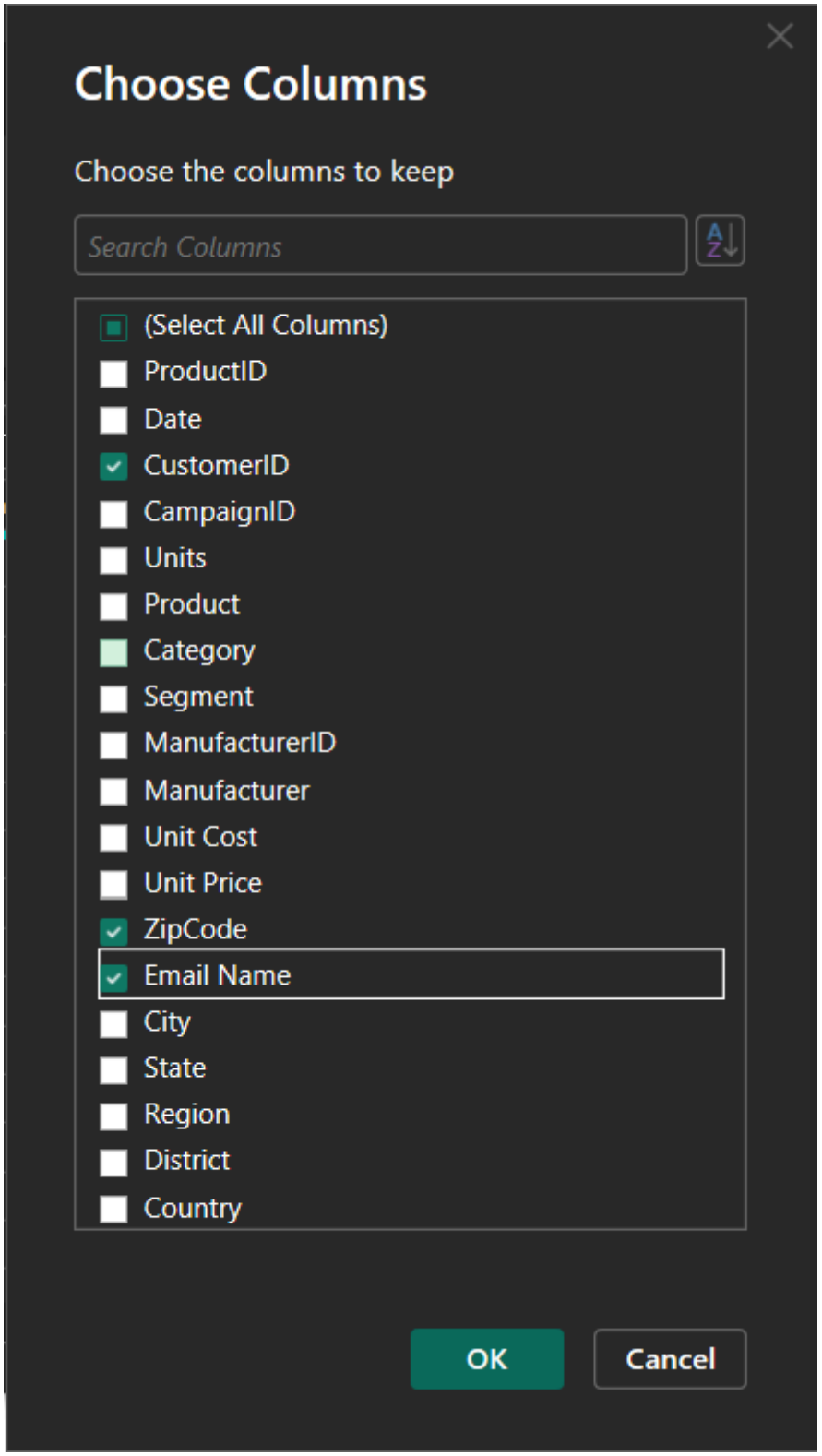


9. Selanjutnya kita pilih semua kolom tabel DimProduct lalu klik kanan dan pilih Remove Duplicate untuk menghilangkan duplikasi data

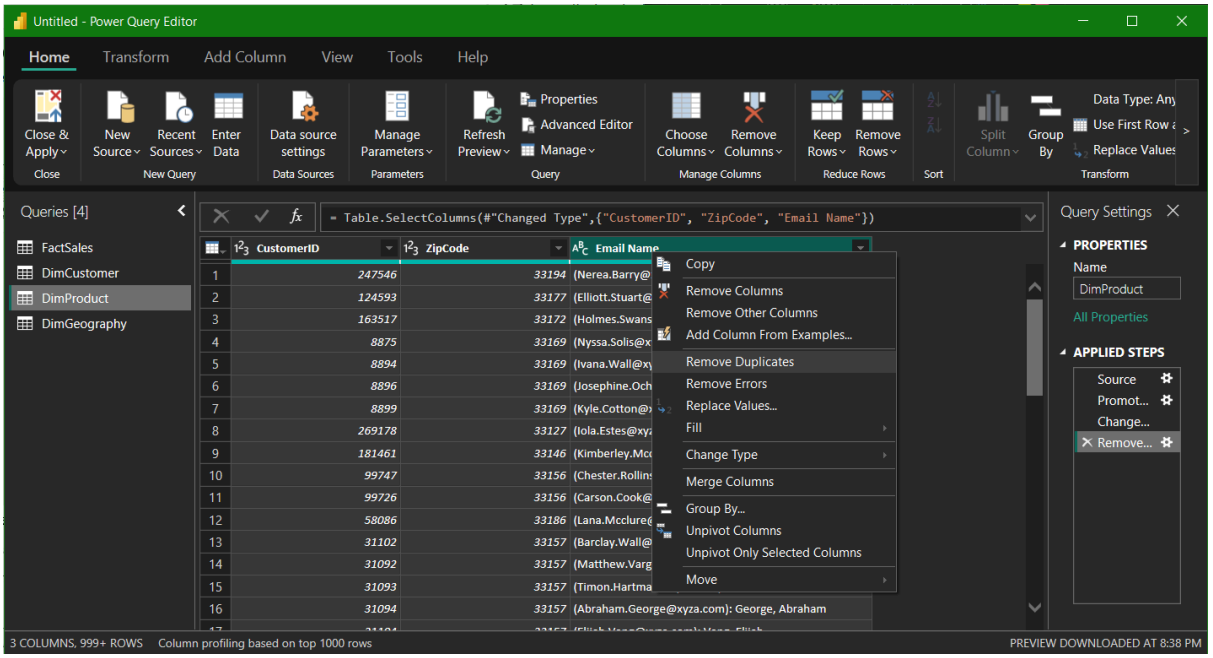

Untitled - Power Query Editor
Home
Transform
Add Column
View
Tools
Help
Close & Apply
New Source
Recent Sources
Enter Data
Data source settings
Manage Parameters
Refresh Preview
Properties
Advanced Editor
Manage
Choose Columns
Remove Columns
Keep Rows
Remove Rows
Split Column
Group By
Data Type: Any
Use First Row a
Replace Values
Close
New Query
Data Sources
Parameters
Query
Manage Columns
Reduce Rows
Sort
Transform
Queries
= Table.SelectColumns(#"Changed Type",{"ProductID", "Product", "Category", "Segment", "Unit Cost", "Unit Price"})
Product
Category
Segment
Unit Cost
Unit Price
Copy
Remove Columns
Remove Other Columns
Add Column From Examples...
Remove Duplicates
Remove Errors
Replace Values...
Fill
Change Type
Merge Columns
Group By...
Unpivot Columns
Unpivot Only Selected Columns
Move
Query Settings
PROPERTIES
Name
DimCustomer
All Properties
APPLIED STEPS
Source
Promot...
Change...
Remove...
6 COLUMNS, 999+ ROWS
Column profiling based on top 1000 rows
PREVIEW DOWNLOADED AT 8:35 PM

10. Setelah bagian DimProduct selesai, kita beralih ke DimCustomer lalu klik Choose Column dan pilih kolom CustomerID, ZipCode, Email Name.

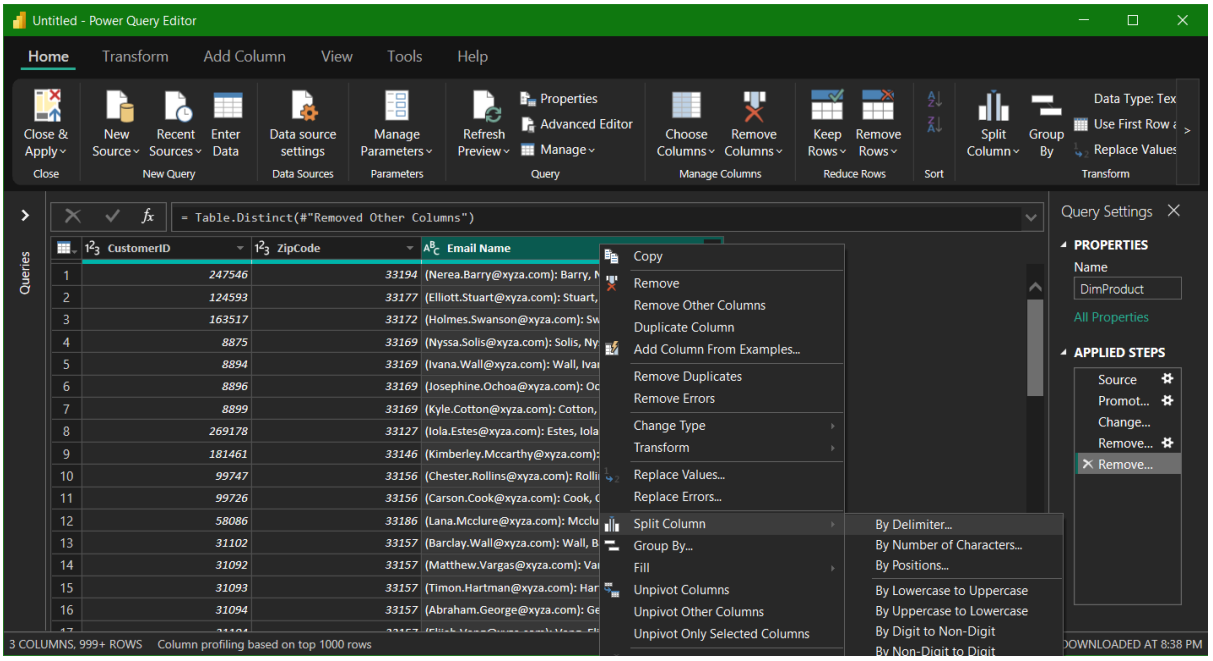


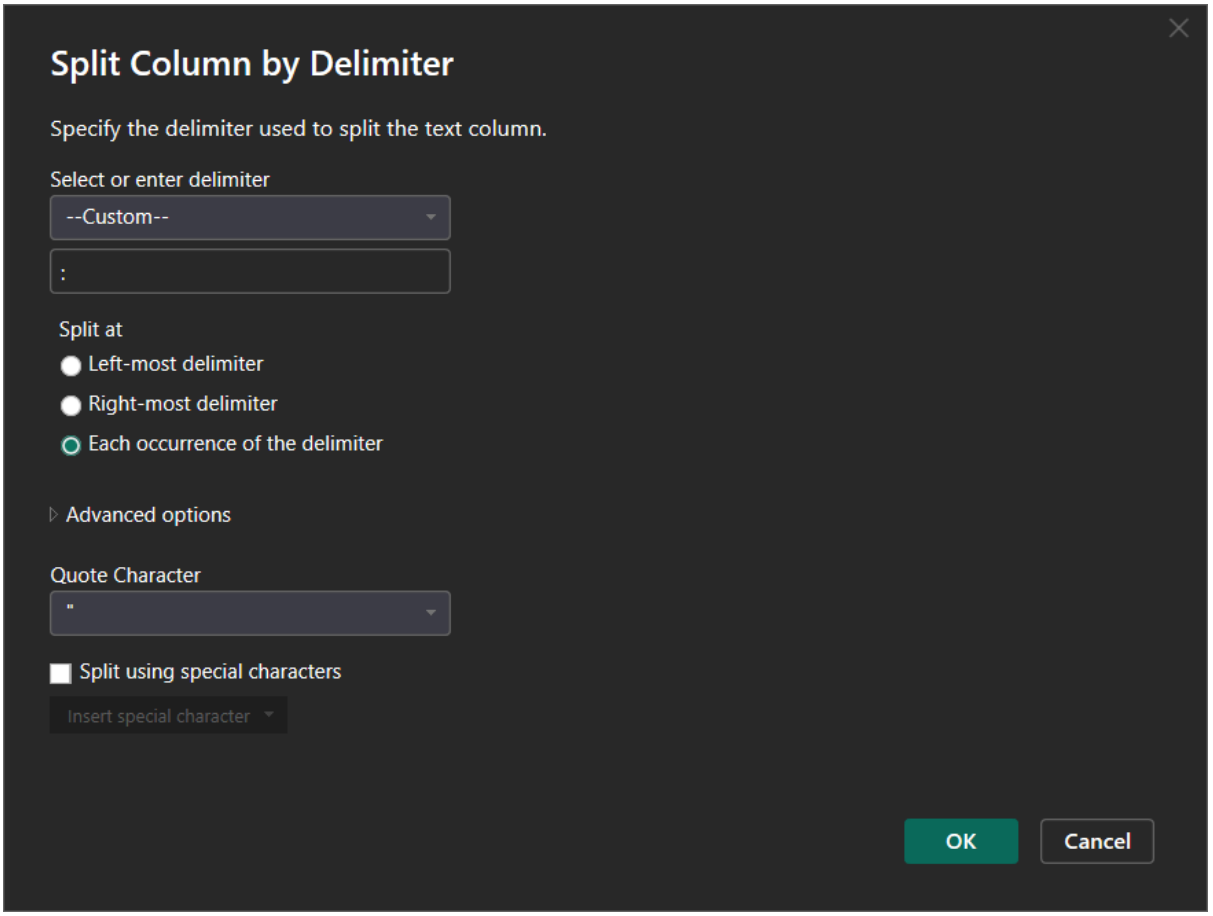


11. Pilih semua kolom DimCustomer lalu klik kanan dan pilih Remove Duplicate

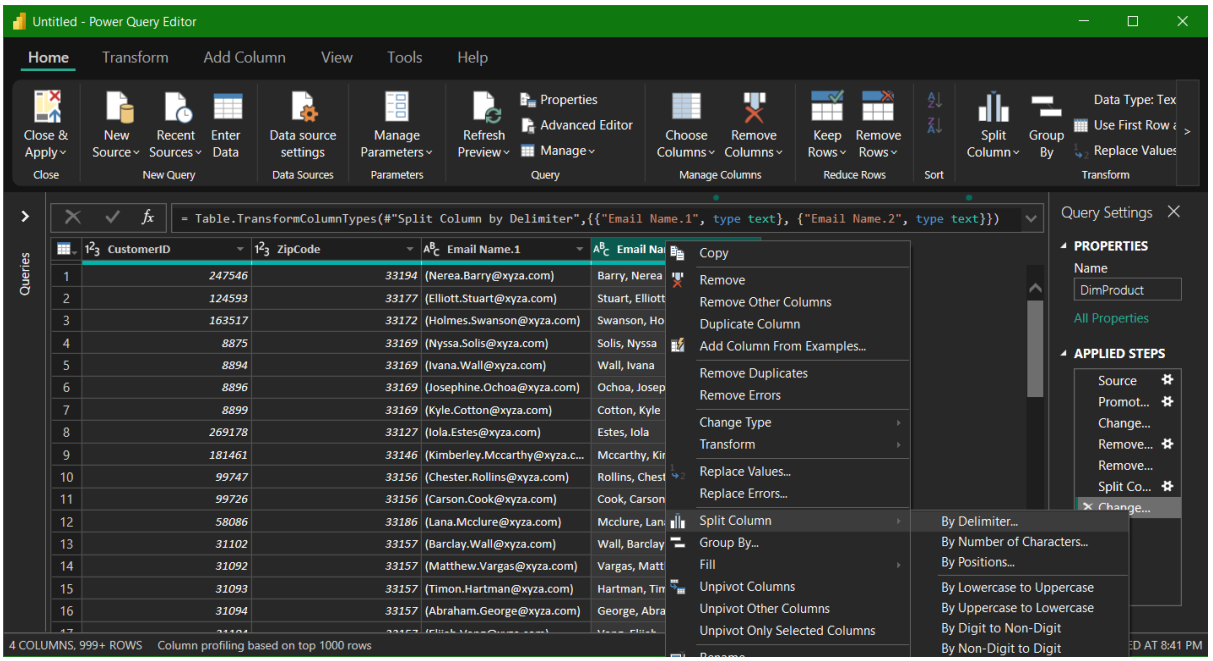


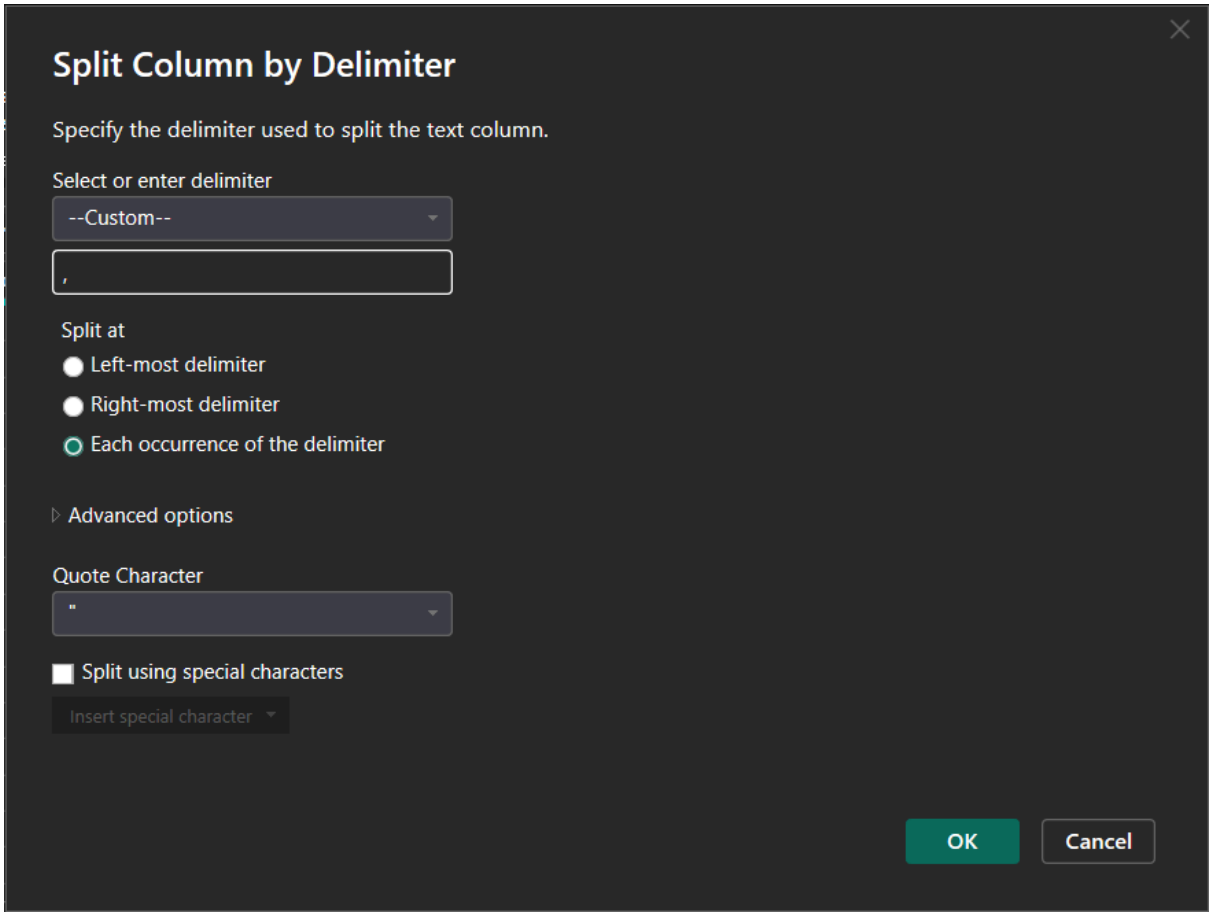
12. Lanjut kita select kolom Email Name klik kanan pilih Split Column > By Delimiter.. untuk memisahkan isi data email dan nama menjadi dua kolom terpisah. Muncul opsi untuk mengatur Delimiternya lalu pilih Custom dan isi dengan delimiter titik 2 dan spasi, lalu klik Ok



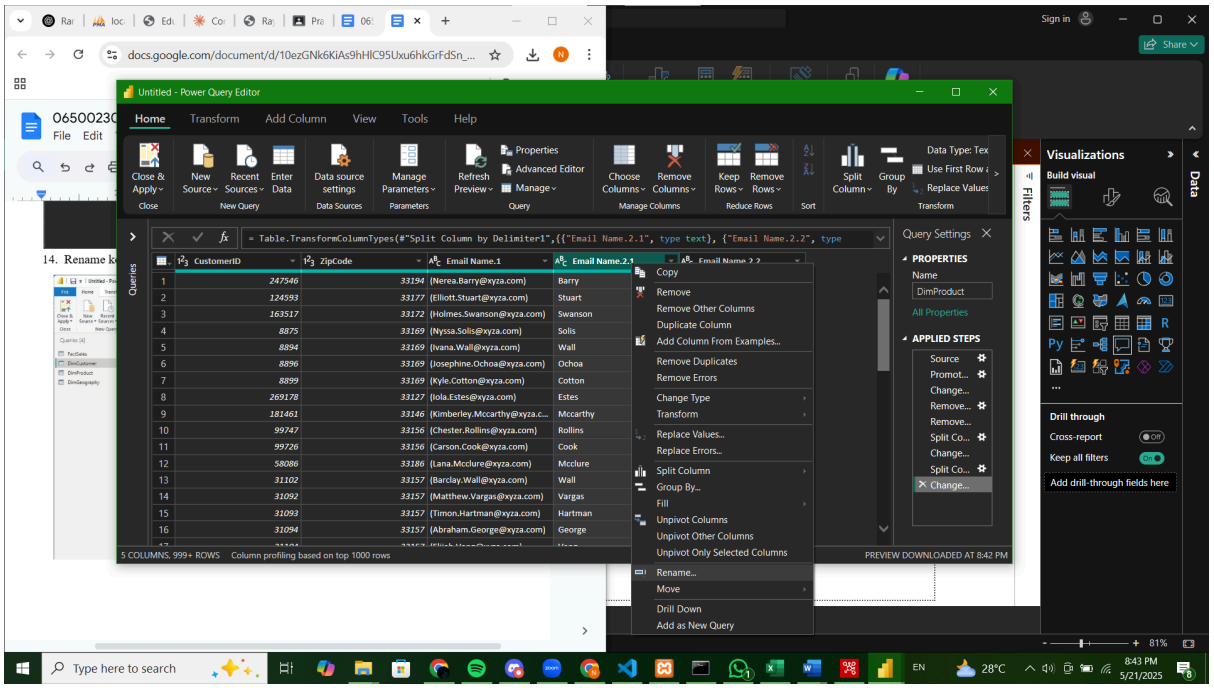


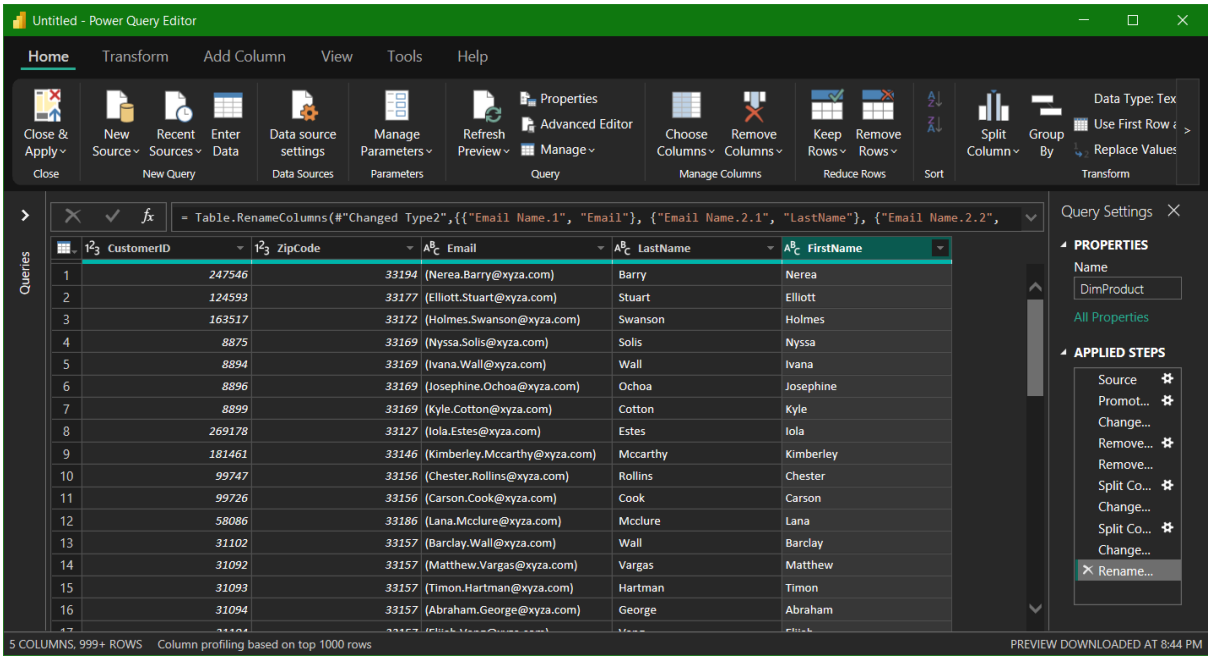
13. Muncul kolom baru hasil Split tadi. Klik kanan lalu Split Column untuk melanjutkan split data nama depan dan nama belakang. Muncul opsi untuk mengatur Delimiternya lalu pilih Custom dan isi dengan delimiter koma spasi, lalu klik Ok





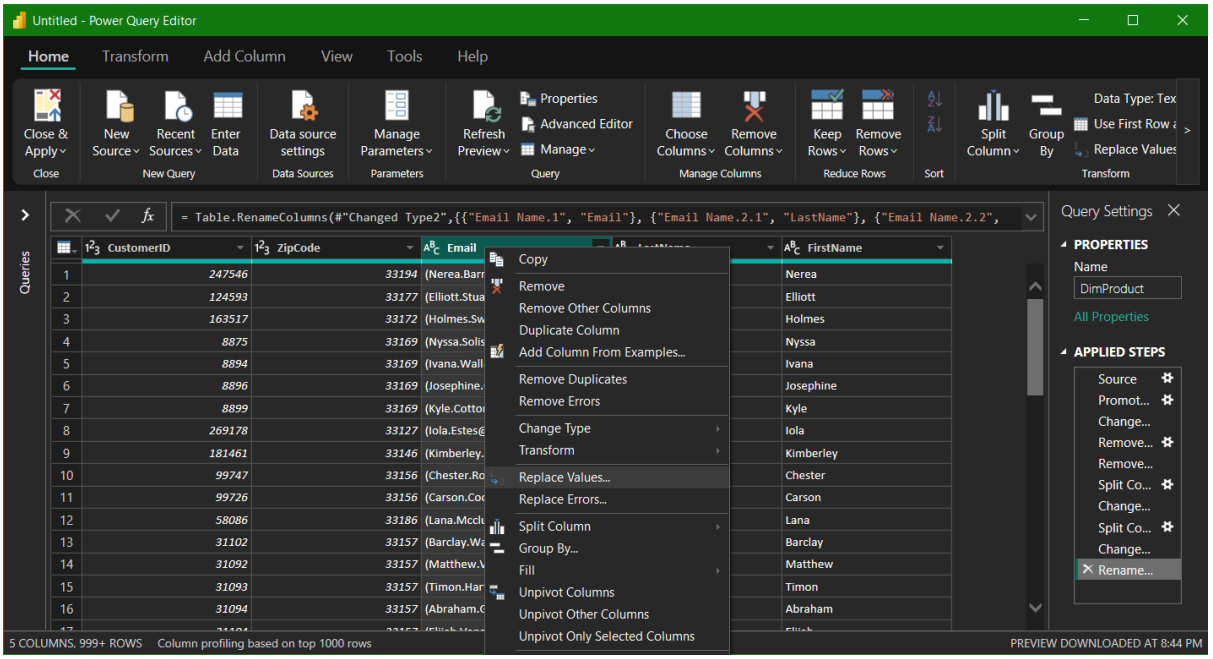
14. Rename kolom hasil Split tadi menjadi Email, LastName, FirstName

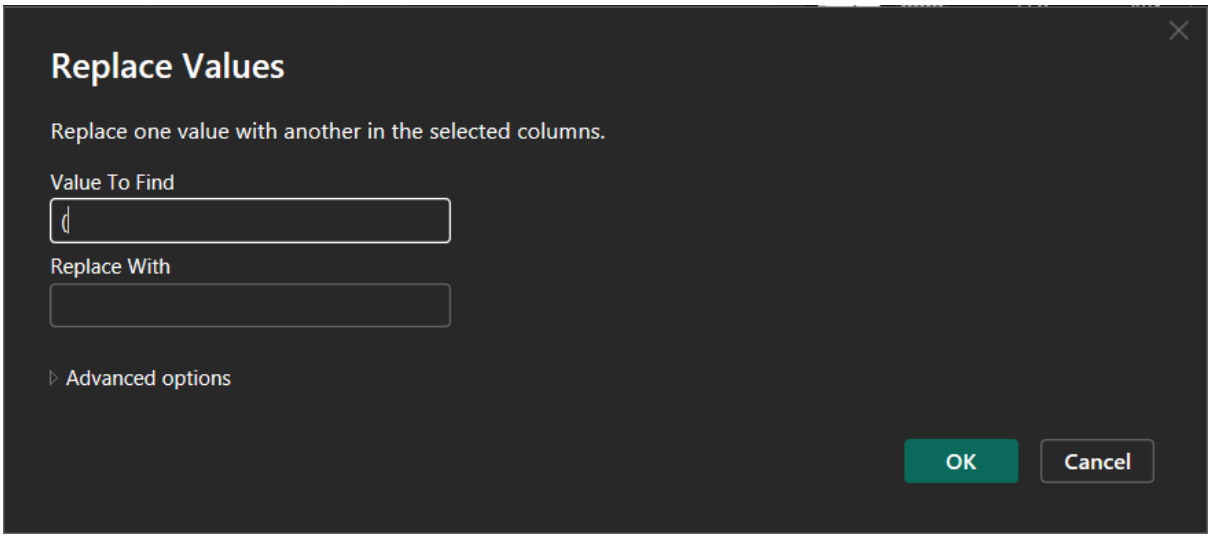


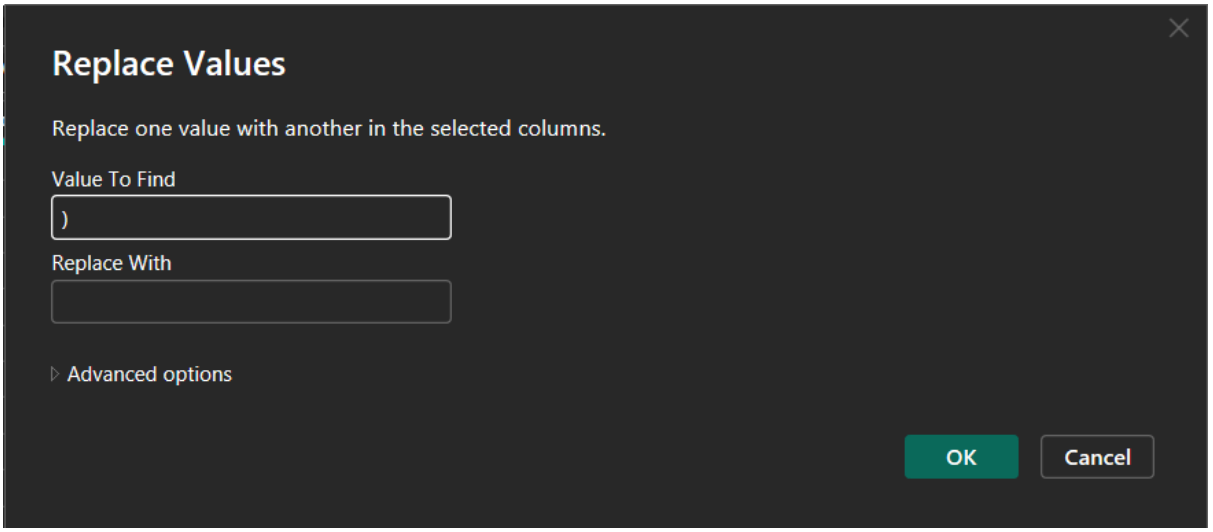


	CustomerID	ZipCode	Email	LastName	FirstName
1	247546	33194	(Nerea.Barry@xyza.com)	Barry	Nerea
2	124593	33177	(Elliott.Stuart@xyza.com)	Stuart	Elliott
3	163517	33172	(Holmes.Swanson@xyza.com)	Swanson	Holmes
4	8875	33169	(Nyssa.Solis@xyza.com)	Solis	Nyssa
5	8894	33169	(Ivana.Wall@xyza.com)	Wall	Ivana
6	8896	33169	(Josephine.Ochoa@xyza.com)	Ochoa	Josephine
7	8899	33169	(Kyle.Cotton@xyza.com)	Cotton	Kyle
8	269178	33127	(Iola.Estes@xyza.com)	Estes	Iola
9	181461	33146	(Kimberley.Mccarthy@xyza.com)	Mccarthy	Kimberley
10	99747	33156	(Chester.Rollins@xyza.com)	Rollins	Chester
11	99726	33156	(Carson.Cook@xyza.com)	Cook	Carson
12	58086	33186	(Lana.Mcclure@xyza.com)	Mcclure	Lana
13	31102	33157	(Barclay.Wall@xyza.com)	Wall	Barclay
14	31092	33157	(Matthew.Vargas@xyza.com)	Vargas	Matthew
15	31093	33157	(Timon.Hartman@xyza.com)	Hartman	Timon
16	31094	33157	(Abraham.George@xyza.com)	George	Abraham

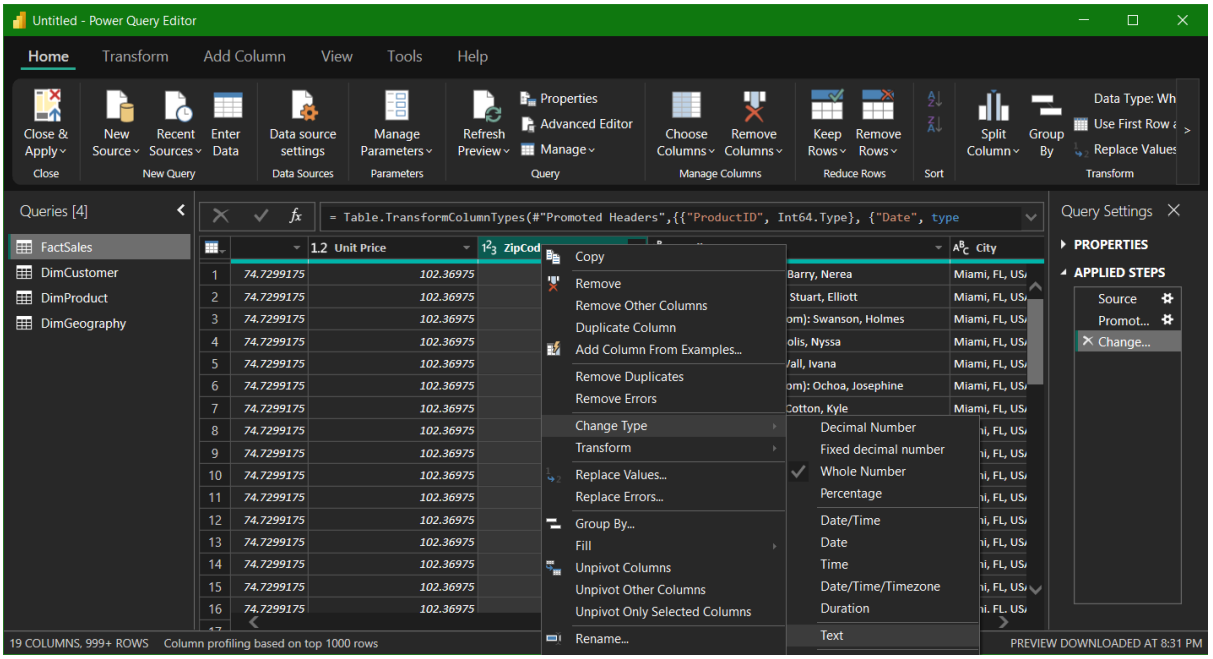
15. Pilih kolom Email, klik kanan Replace Values untuk menghilangkan tanda kurung ( ) di data tersebut. Lakukan step ini 2x dengan mengisi Value To Find pertama dengan (lalu yang kedua dengan) dan Replace with kosongkan.

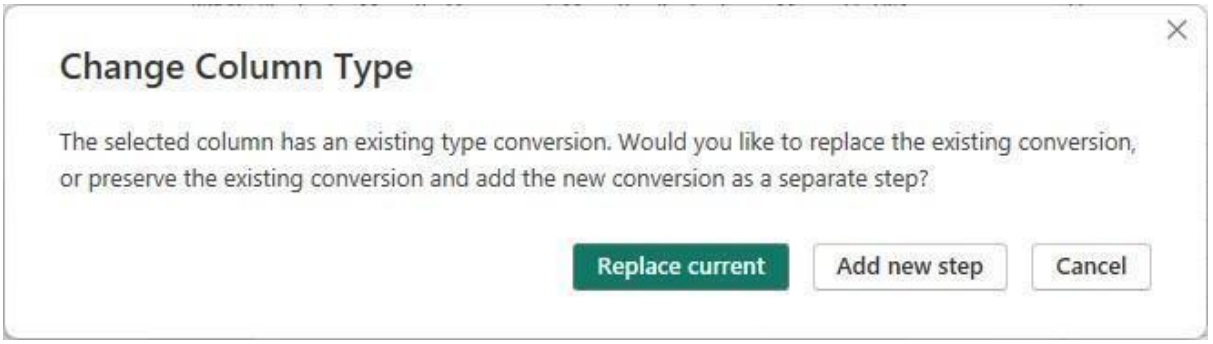




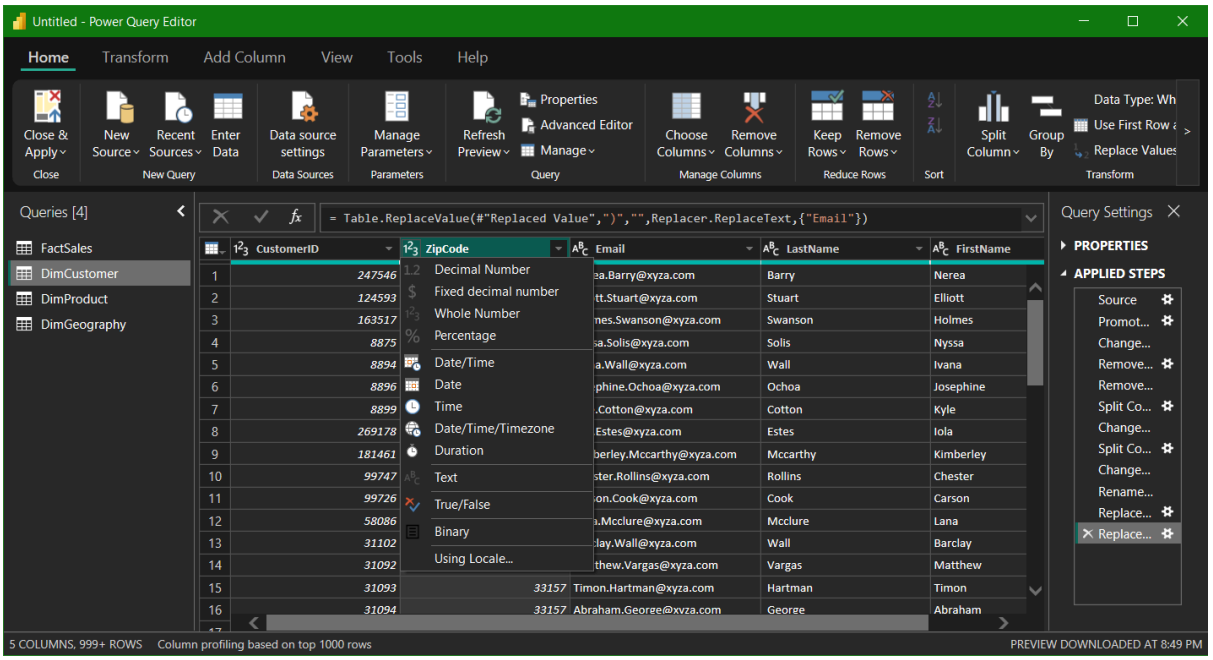


16. Kita kembali ke FactTabel untuk mengubah tipe data di kolom ZipCode menjadi Text dengan klik kanan pada kolom lalu pilih Change Type > Text. (Tips: kita juga bisa langsung klik icon samping nama kolom untuk merubah tipe kolom data). Akan muncul notifikasi untuk Change Column Type lalu klik Replace current

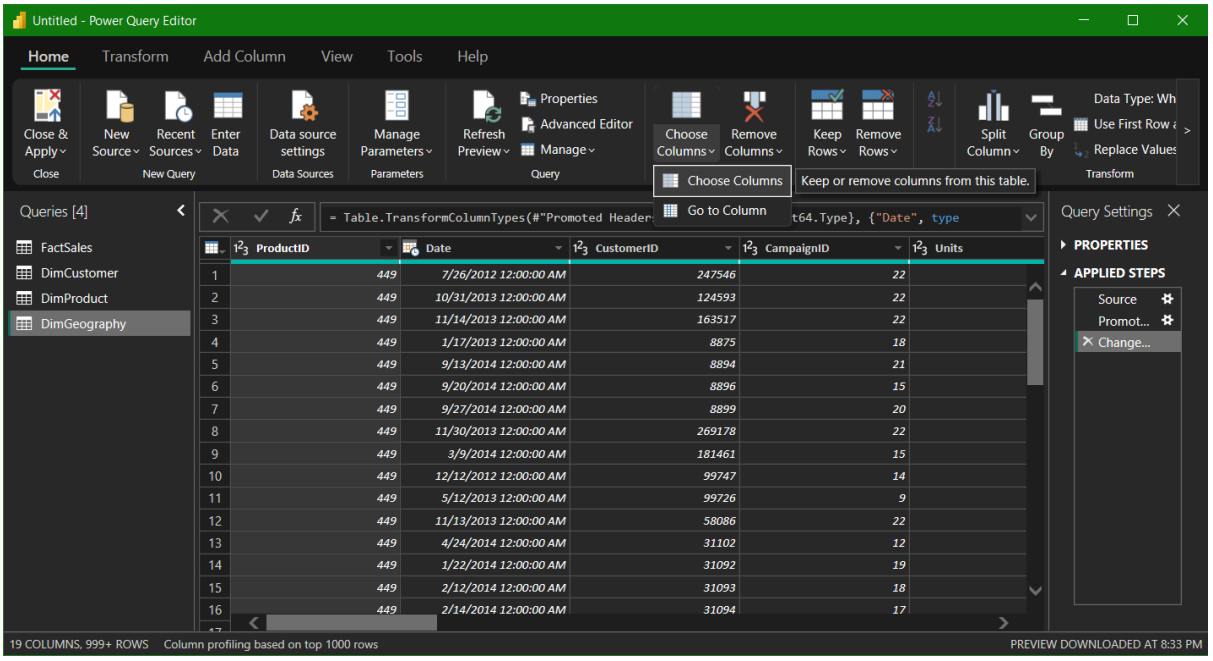


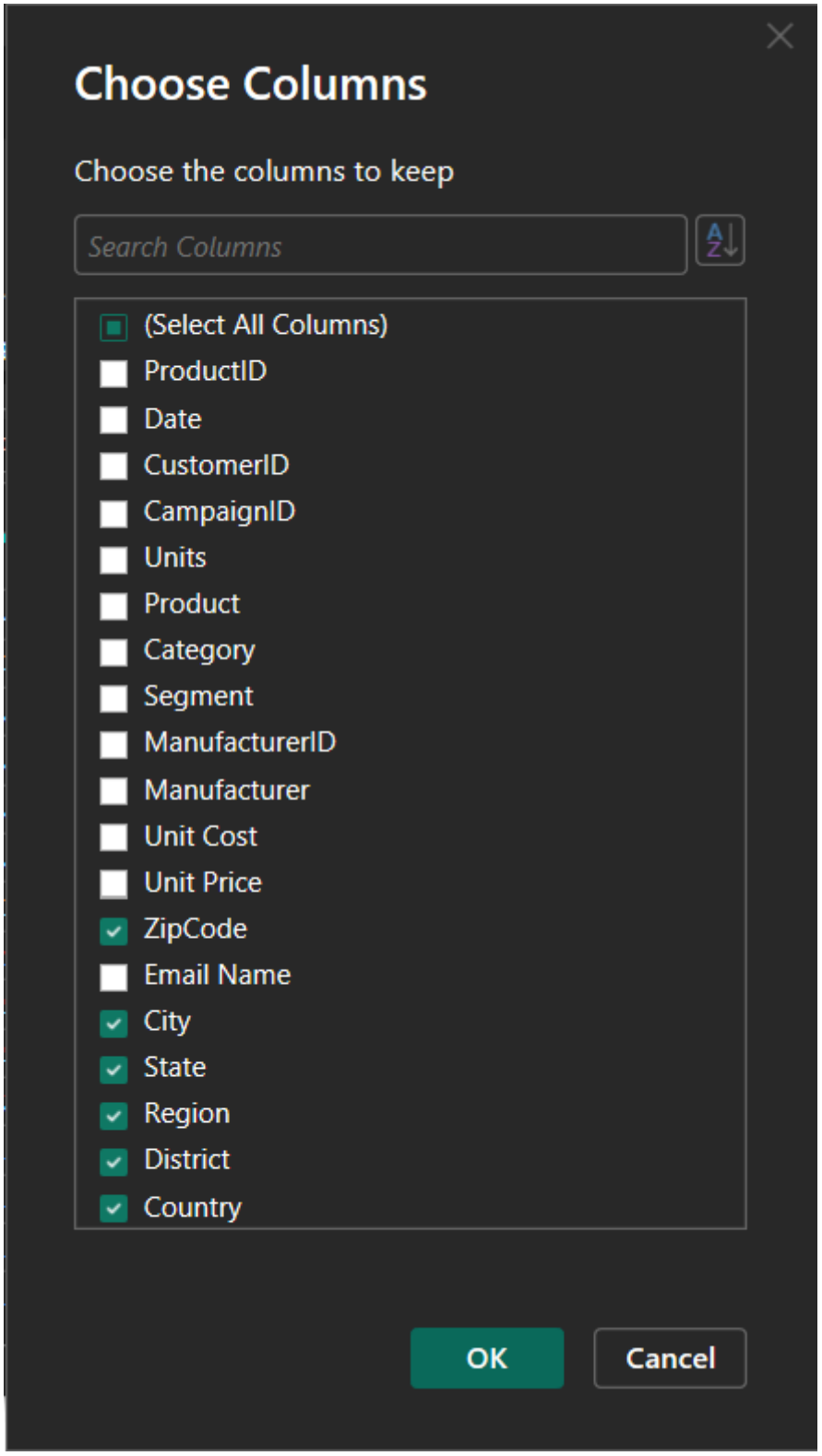

Change Column Type
The selected column has an existing type conversion. Would you like to replace the existing conversion, or preserve the existing conversion and add the new conversion as a separate step?
Replace current
Add new step
Cancel

17. Beralih ke DimCustomer untuk merubah tipe kolom data ZipCode apabila tidak ikut berubah menjadi Text pada step sebelumnya

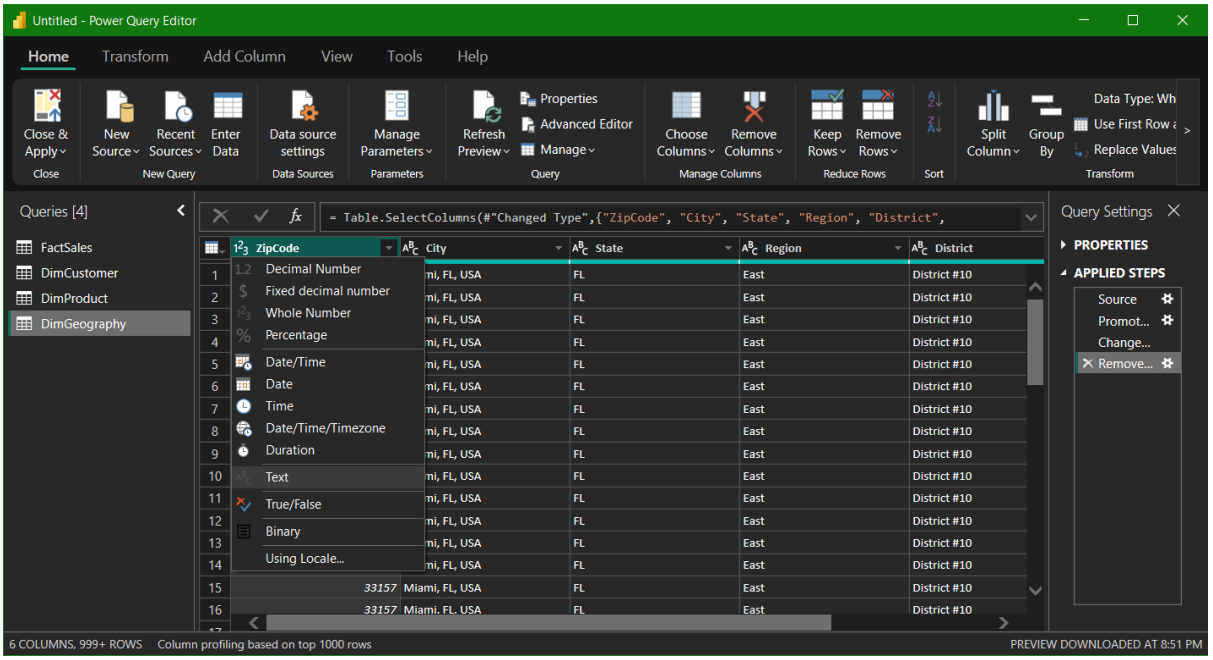


18. Selanjutnya kita beralih ke DimGeography lalu klik Choose Column lalu pilih kolom ZipCode, City, State, Region, District, Country.





19. Lalu ubah tipe kolom data ZipCode DimGeography menjadi Text


Untitled - Power Query Editor
Home
Transform
Add Column
View
Tools
Help
Close & Apply
New Source
Recent Sources
Enter Data
Data source settings
Manage Parameters
Refresh Preview
Properties
Advanced Editor
Manage
Choose Columns
Remove Columns
Keep Rows
Remove Rows
Split Column
Group By
Data Type: Wh
Use First Row a
Replace Values
Close
New Query
Data Sources
Parameters
Query
Manage Columns
Reduce Rows
Sort
Transform
Queries [4]
FactSales
DimCustomer
DimProduct
DimGeography
= Table.SelectColumns(#"Changed Type",{"ZipCode", "City", "State", "Region", "District",
ZipCode
City
State
Region
District
Decimal Number
Fixed decimal number
Whole Number
Percentage
Date/Time
Date
Time
Date/Time/Timezone
Duration
Text
True/False
Binary
Using Locale...
Query Settings
PROPERTIES
APPLIED STEPS
Source
Promot...
Change...
Remove...
6 COLUMNS, 999+ ROWS
Column profiling based on top 1000 rows
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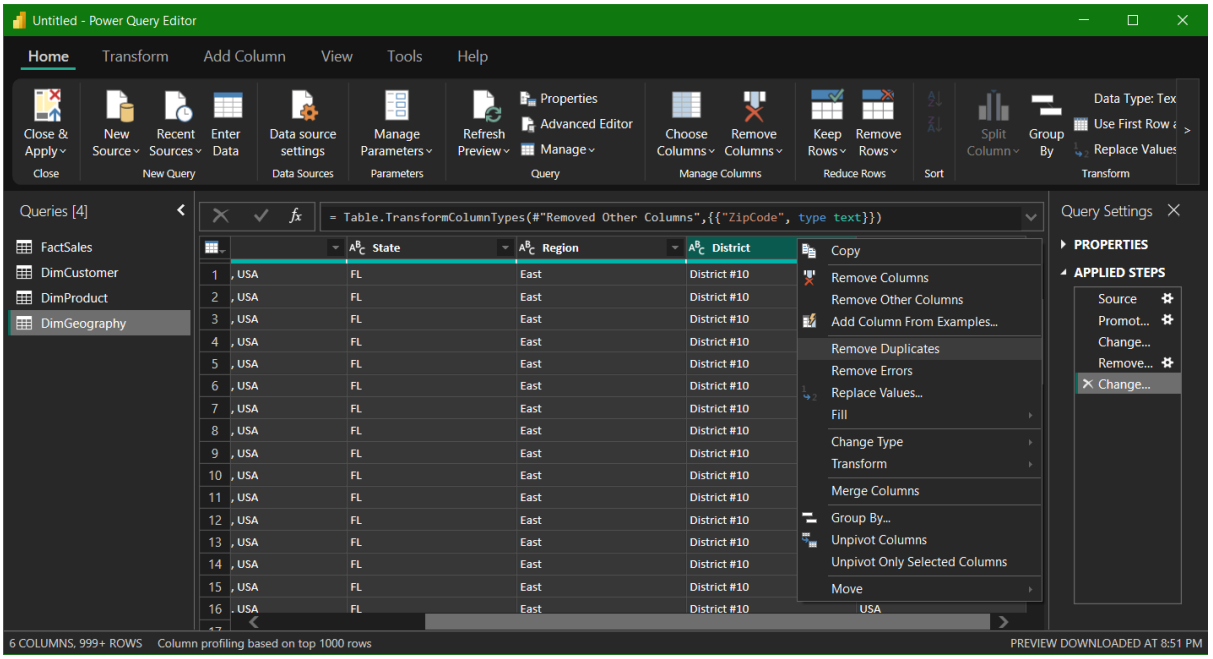
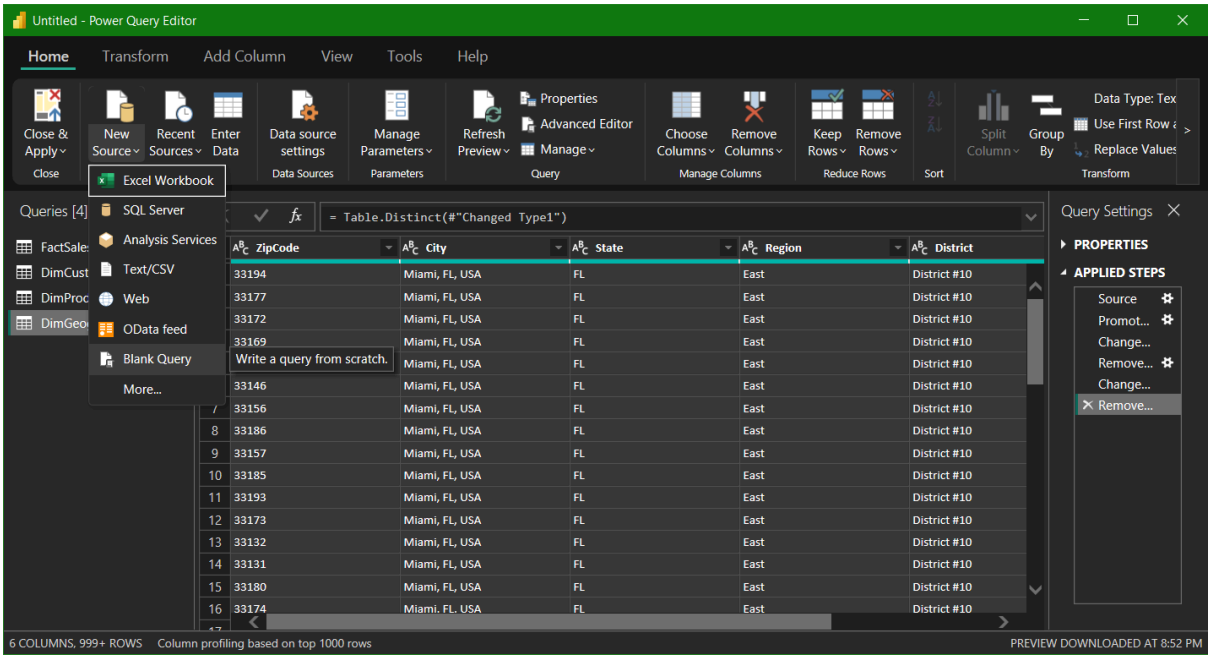
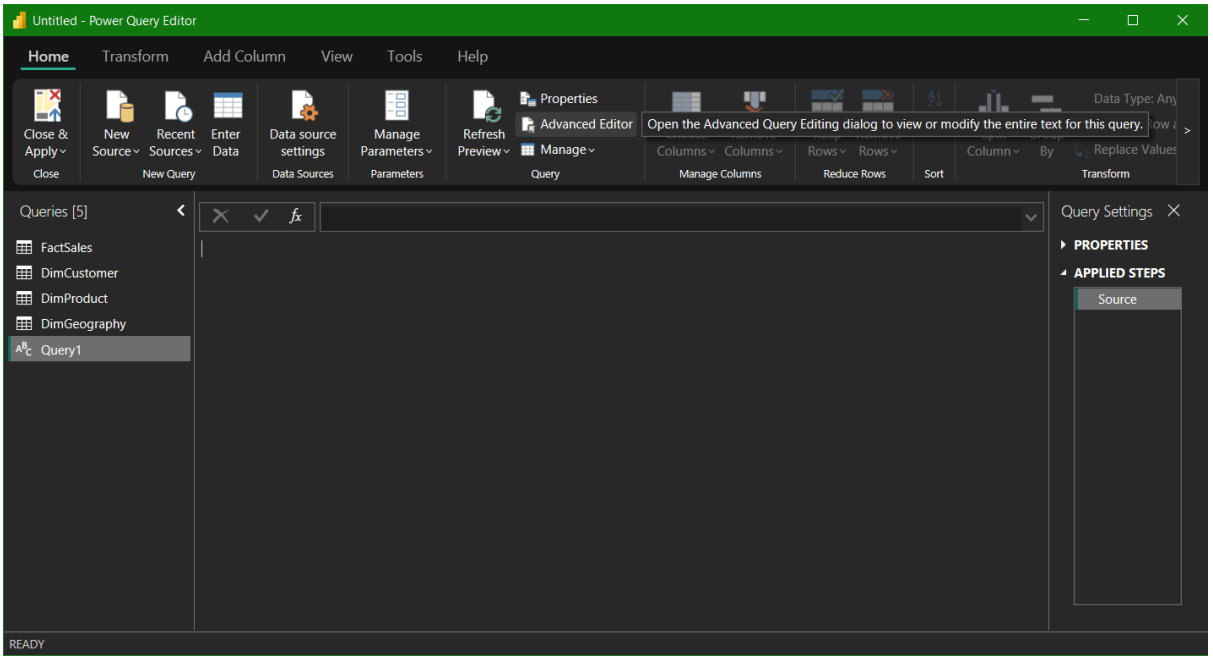
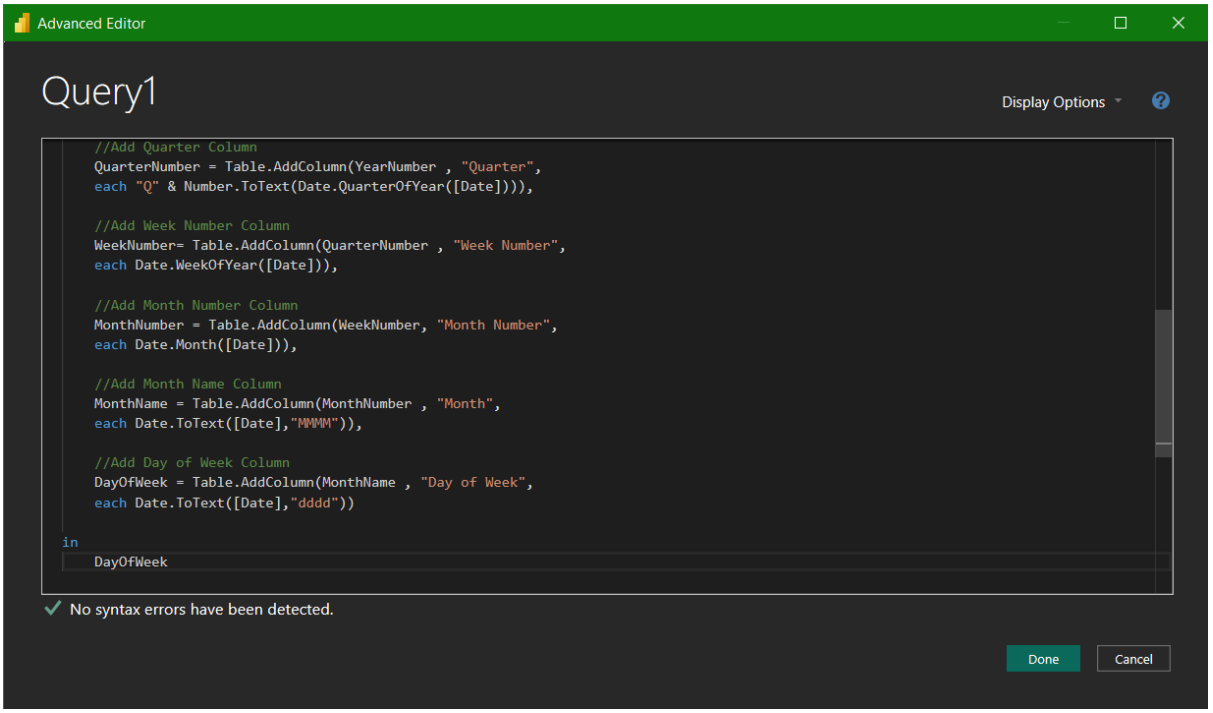
20. Lanjut kita hilangkan duplikasi data pada seluruh kolom dengan Remove Duplicate



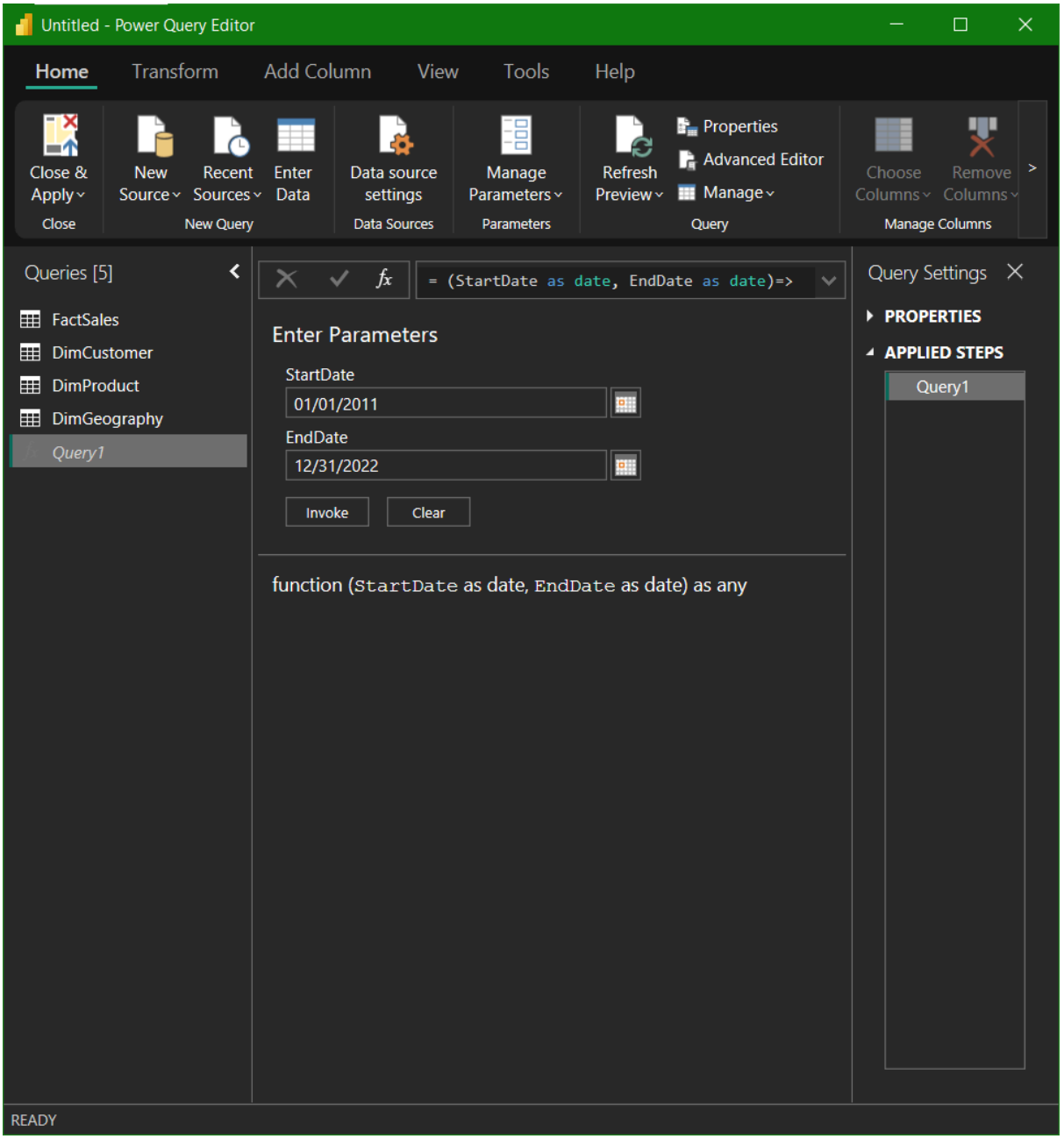
21. Setelah semua selesai, kita perlu membuat DimDate dengan contoh kasus kita perlu seluruh Date agar bisa menampilkan date tertentu. Pertama kita opy kode untuk membuat kostumisasi date di website ini https://devinknightsql.com/2015/06/16/creating-a-date-dimension-with-power-query/ lalu pilih New Source > Blank Query kemudian muncul Query1 lalu klik Advanced Editor lalu Paste kode tadi klik Done.



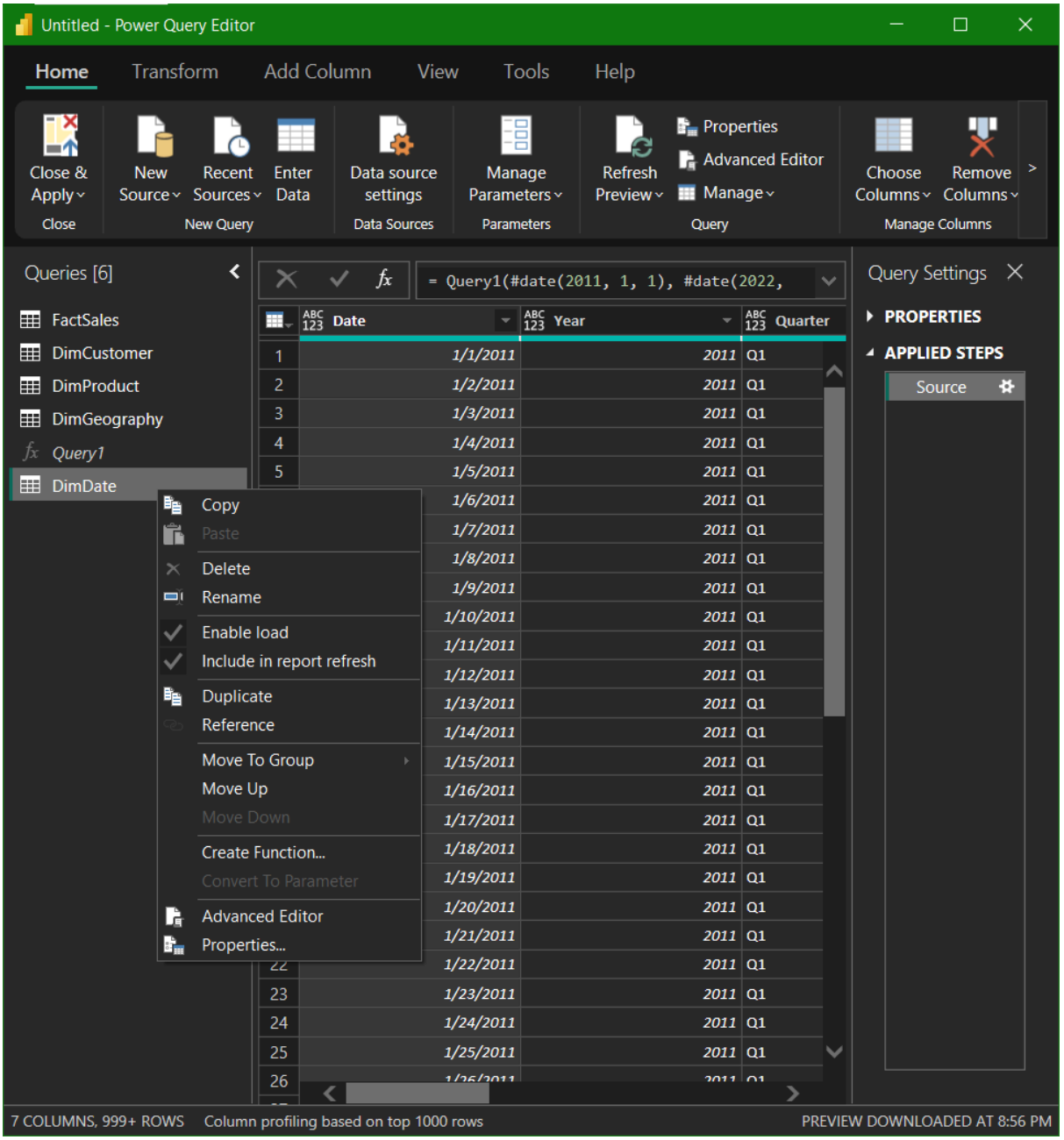

Untitled - Power Query Editor
Home
Transform
Add Column
View
Tools
Help
Close & Apply
New Source
Recent Sources
Enter Data
Data source settings
Manage Parameters
Refresh Preview
Properties
Advanced Editor
Manage
Open the Advanced Query Editing dialog to view or modify the entire text for this query.
Close
New Query
Data Sources
Parameters
Query
Manage Columns
Reduce Rows
Sort
Transform
Queries [5]
FactSales
DimCustomer
DimProduct
DimGeography
Query1
Query Settings
PROPERTIES
APPLIED STEPS
Source
READY

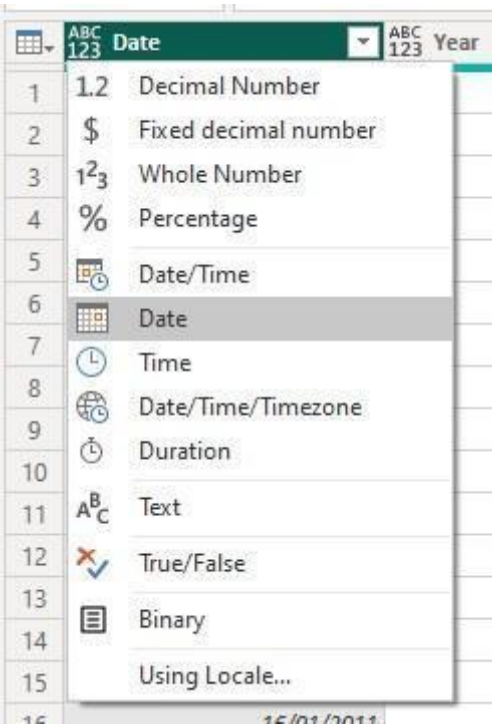

Advanced Editor
Query1
Display Options
//Add Quarter Column
QuarterNumber = Table.AddColumn(YearNumber , "Quarter",
each "Q" & Number.ToText(Date.QuarterOfYear([Date])),
//Add Week Number Column
WeekNumber= Table.AddColumn(QuarterNumber , "Week Number",
each Date.WeekOfYear([Date])),
//Add Month Number Column
MonthNumber = Table.AddColumn(WeekNumber, "Month Number",
each Date.Month([Date])),
//Add Month Name Column
MonthName = Table.AddColumn(MonthNumber , "Month",
each Date.ToText([Date],"MMMM")),
//Add Day of Week Column
DayOfWeek = Table.AddColumn(MonthName , "Day of Week",
each Date.ToText([Date],"dddd"))
in
DayOfWeek
No syntax errors have been detected.
Done
Cancel

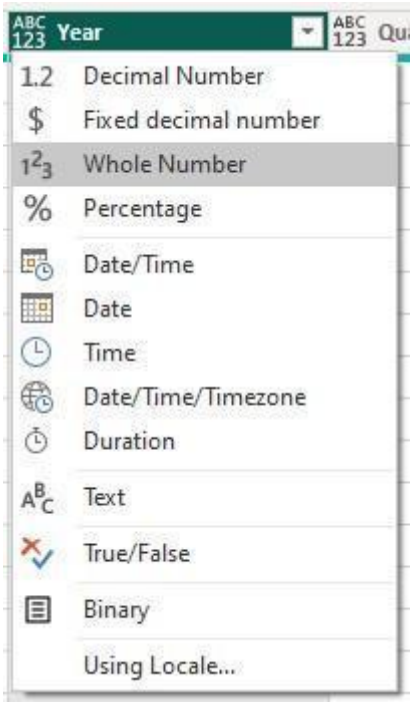
22. Setelah terbuat, masukan Date yang akan kita buat

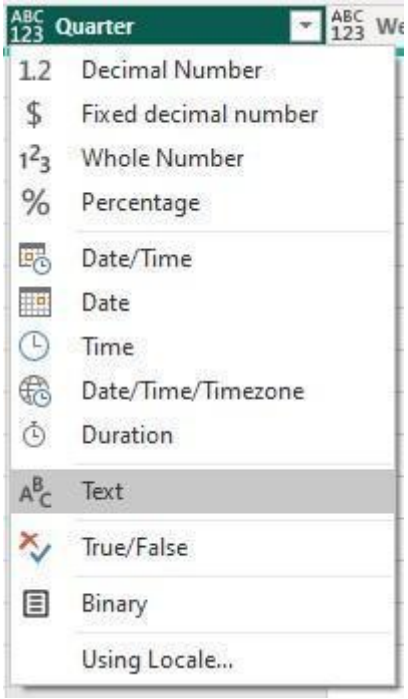


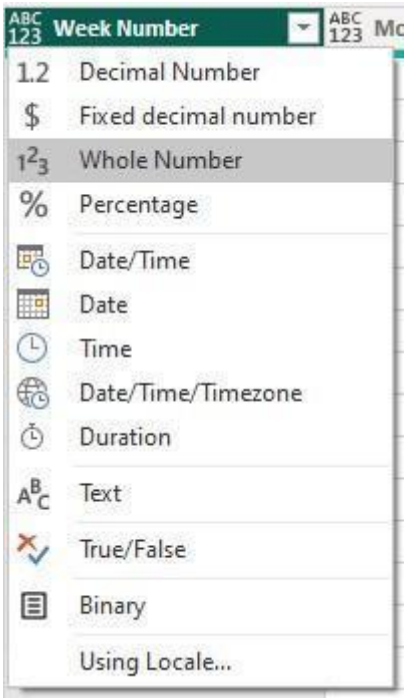
23. Muncul tabel baru bernama Invoked Function lalu kita rename menjadi DimDate dan ubah tipe data tiap kolom seperti screenshoot dibawah ini

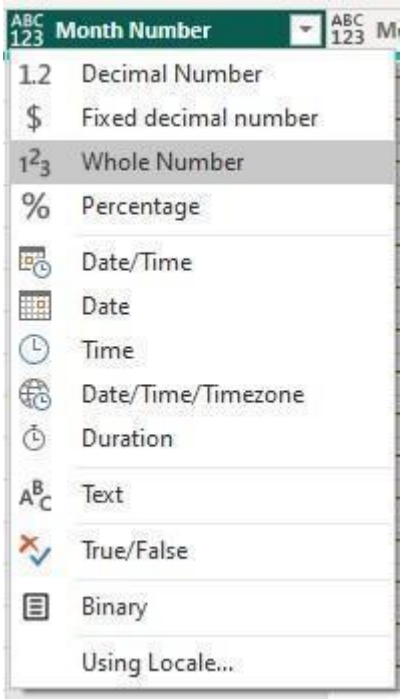

Untitled - Power Query Editor
Home
Transform
Add Column
View
Tools
Help
Close & Apply
New Source
Recent Sources
Enter Data
Data source settings
Manage Parameters
Refresh Preview
Properties
Advanced Editor
Manage
Choose Columns
Remove Columns
Close
New Query
Data Sources
Parameters
Query
Manage Columns
Queries [6]
FactSales
DimCustomer
DimProduct
DimGeography
Query1
DimDate
Copy
Paste
Delete
Rename
Enable load
Include in report refresh
Duplicate
Reference
Move To Group
Move Up
Move Down
Create Function...
Convert To Parameter
Advanced Editor
Properties...
= Query1(#date(2011, 1, 1), #date(2022,
Date
Year
Quarter
Query Settings
PROPERTIES
APPLIED STEPS
Source
7 COLUMNS, 999+ ROWS
Column profiling based on top 1000 rows
PREVIEW DOWNLOADED AT 8:56 PM

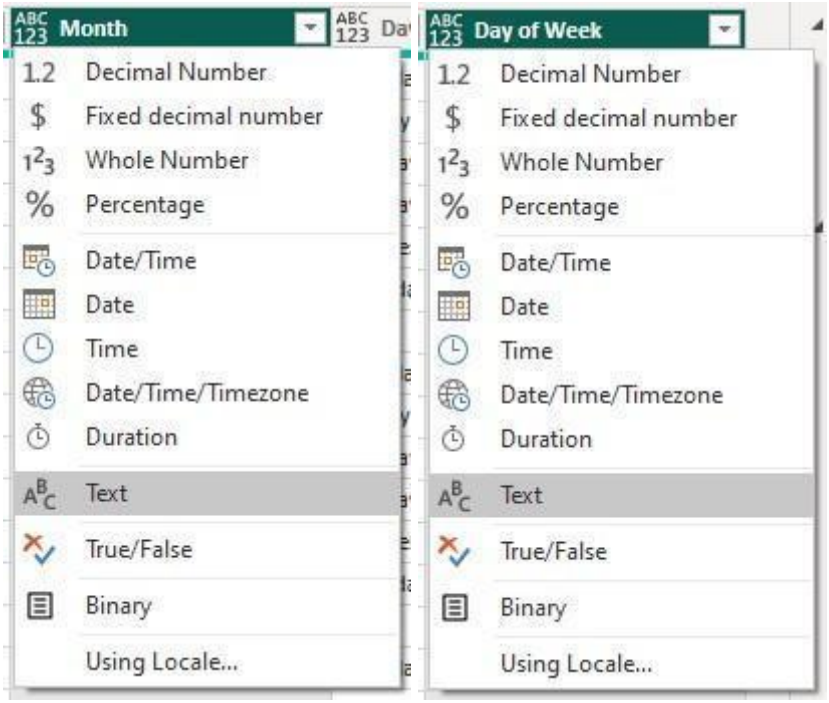

Date
Year
Decimal Number
Fixed decimal number
Whole Number
Percentage
Date/Time
Date
Time
Date/Time/Timezone
Duration
Text
True/False
Binary
Using Locale...

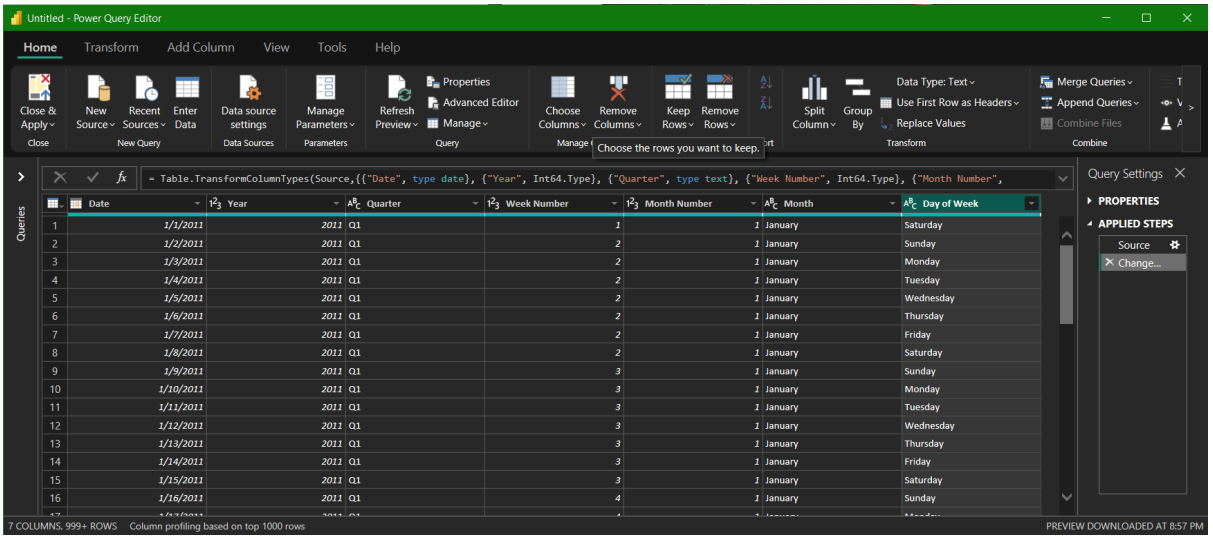

Year
Decimal Number
Fixed decimal number
Whole Number
Percentage
Date/Time
Date
Time
Date/Time/Timezone
Duration
Text
True/False
Binary
Using Locale...


Quarter
Decimal Number
Fixed decimal number
Whole Number
Percentage
Date/Time
Date
Time
Date/Time/Timezone
Duration
Text
True/False
Binary
Using Locale...

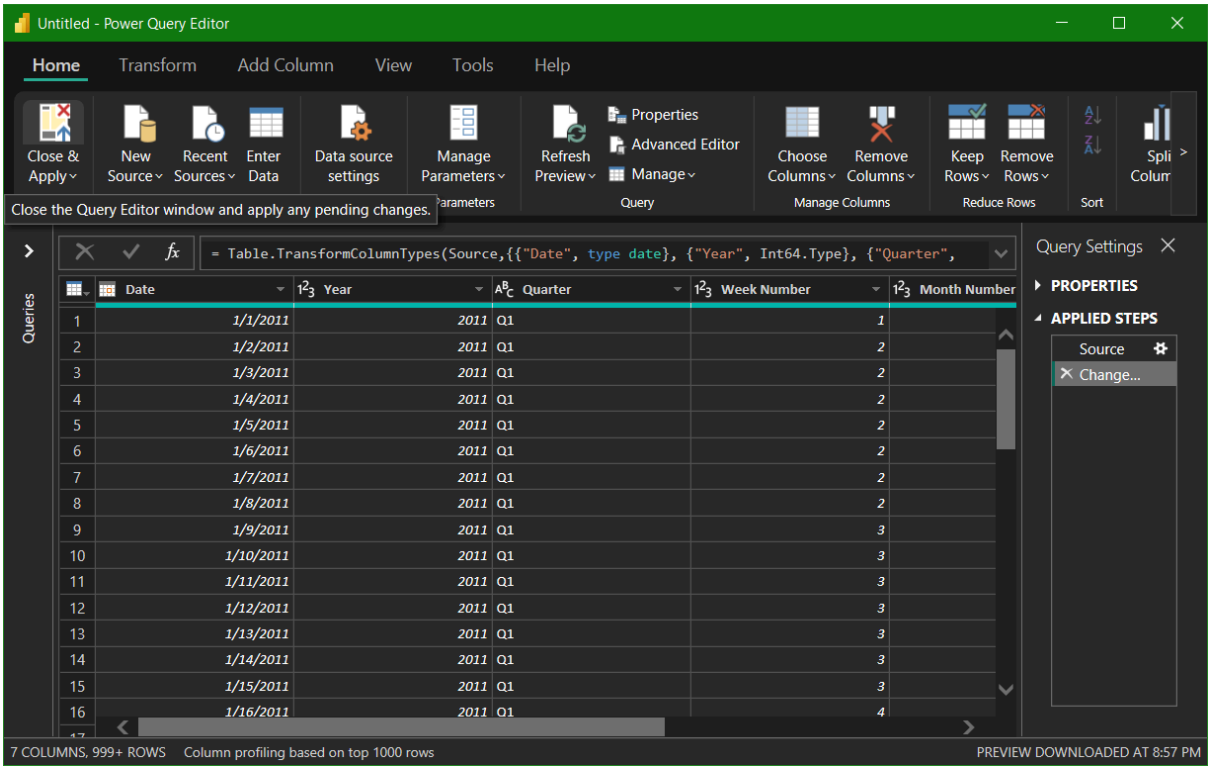

Week Number
Decimal Number
Fixed decimal number
Whole Number
Percentage
Date/Time
Date
Time
Date/Time/Timezone
Duration
Text
True/False
Binary
Using Locale...


Month Number
Decimal Number
Fixed decimal number
Whole Number
Percentage
Date/Time
Date
Time
Date/Time/Timezone
Duration
Text
True/False
Binary
Using Locale...



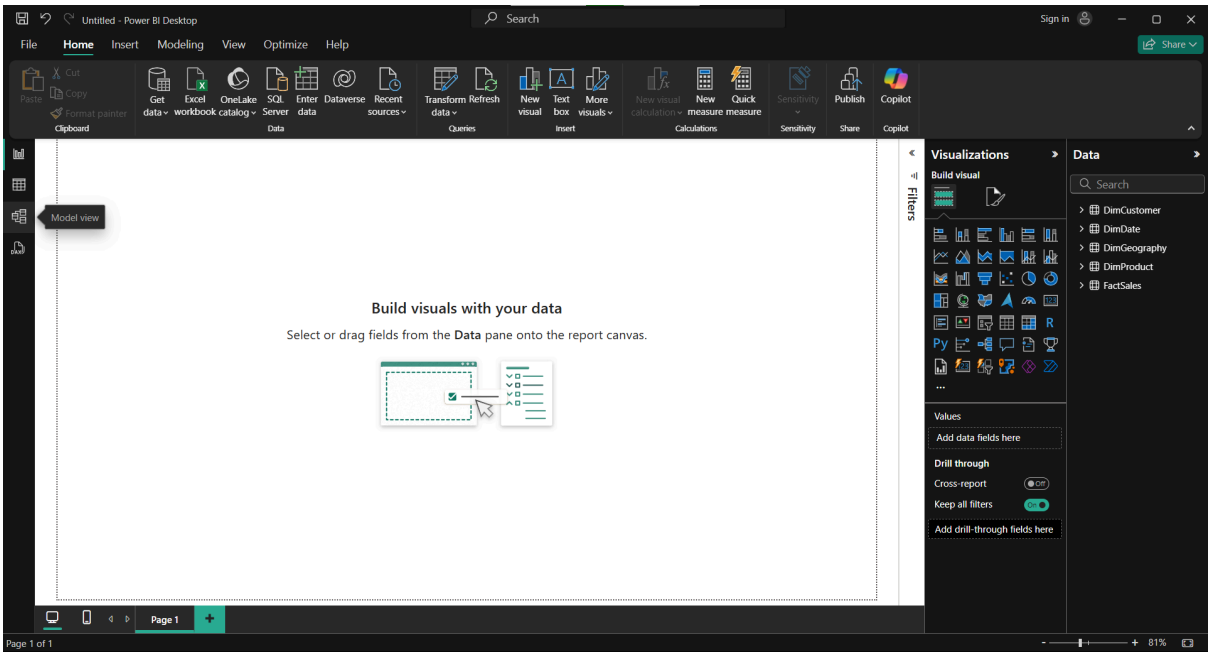


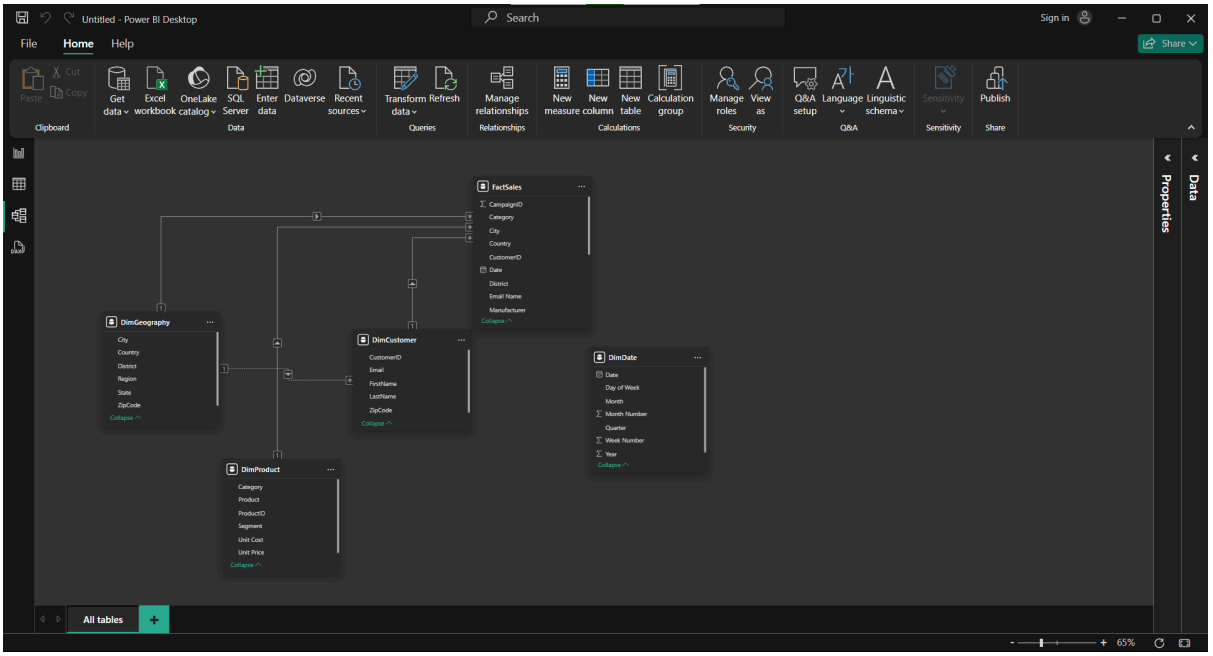
24. Setelah semua step sebelumnya sudah dilakukan, kita cek hasil sementara dengan klik Close & Apply



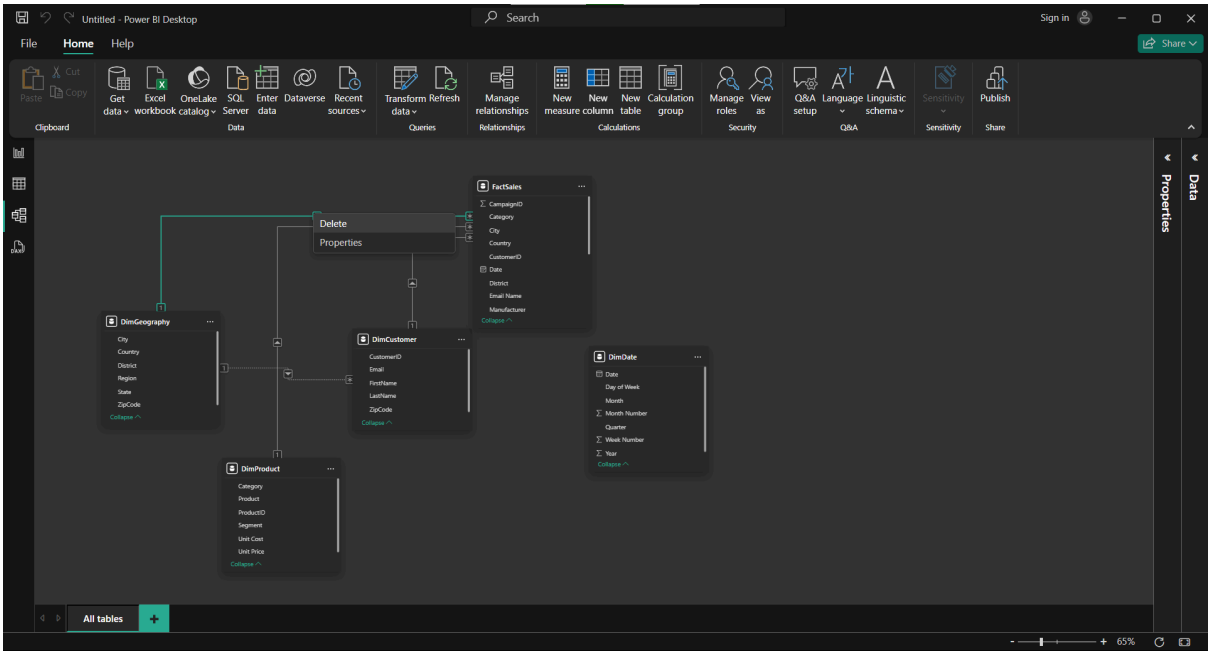
	Date	Year	Quarter	Week Number	Month Number
1	1/1/2011	2011	Q1	1	
2	1/2/2011	2011	Q1	2	
3	1/3/2011	2011	Q1	2	
4	1/4/2011	2011	Q1	2	
5	1/5/2011	2011	Q1	2	
6	1/6/2011	2011	Q1	2	
7	1/7/2011	2011	Q1	2	
8	1/8/2011	2011	Q1	2	
9	1/9/2011	2011	Q1	3	
10	1/10/2011	2011	Q1	3	
11	1/11/2011	2011	Q1	3	
12	1/12/2011	2011	Q1	3	
13	1/13/2011	2011	Q1	3	
14	1/14/2011	2011	Q1	3	
15	1/15/2011	2011	Q1	3	
16	1/16/2011	2011	Q1	4	

25. Kita beralih ke Model View

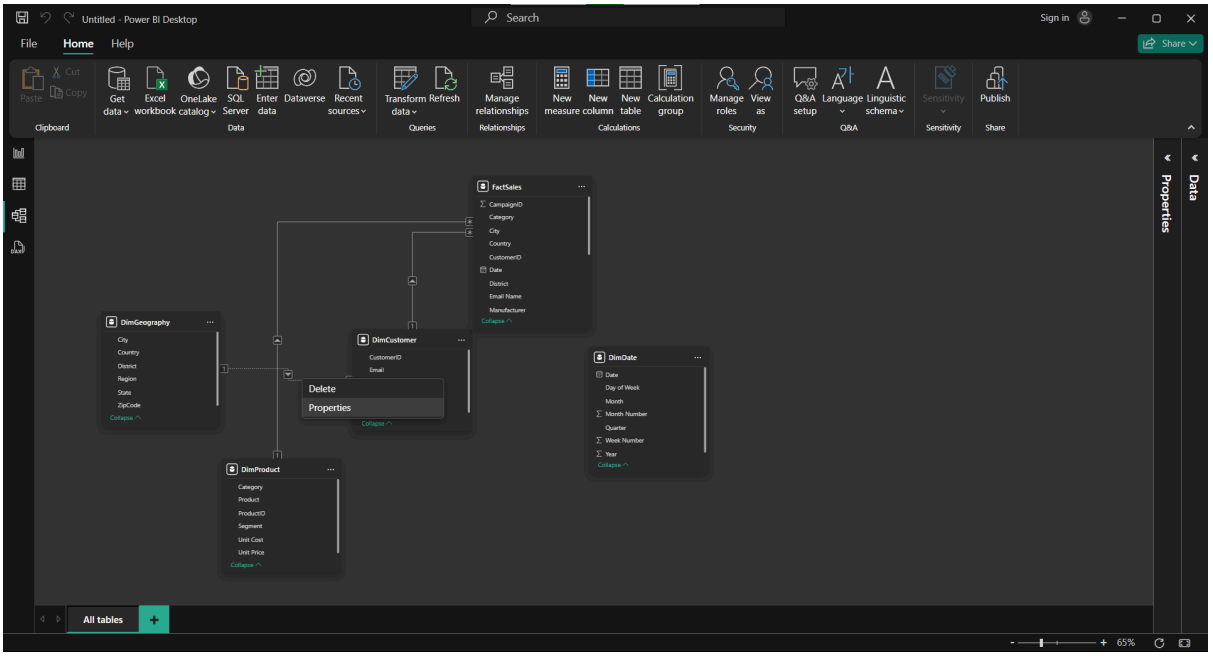


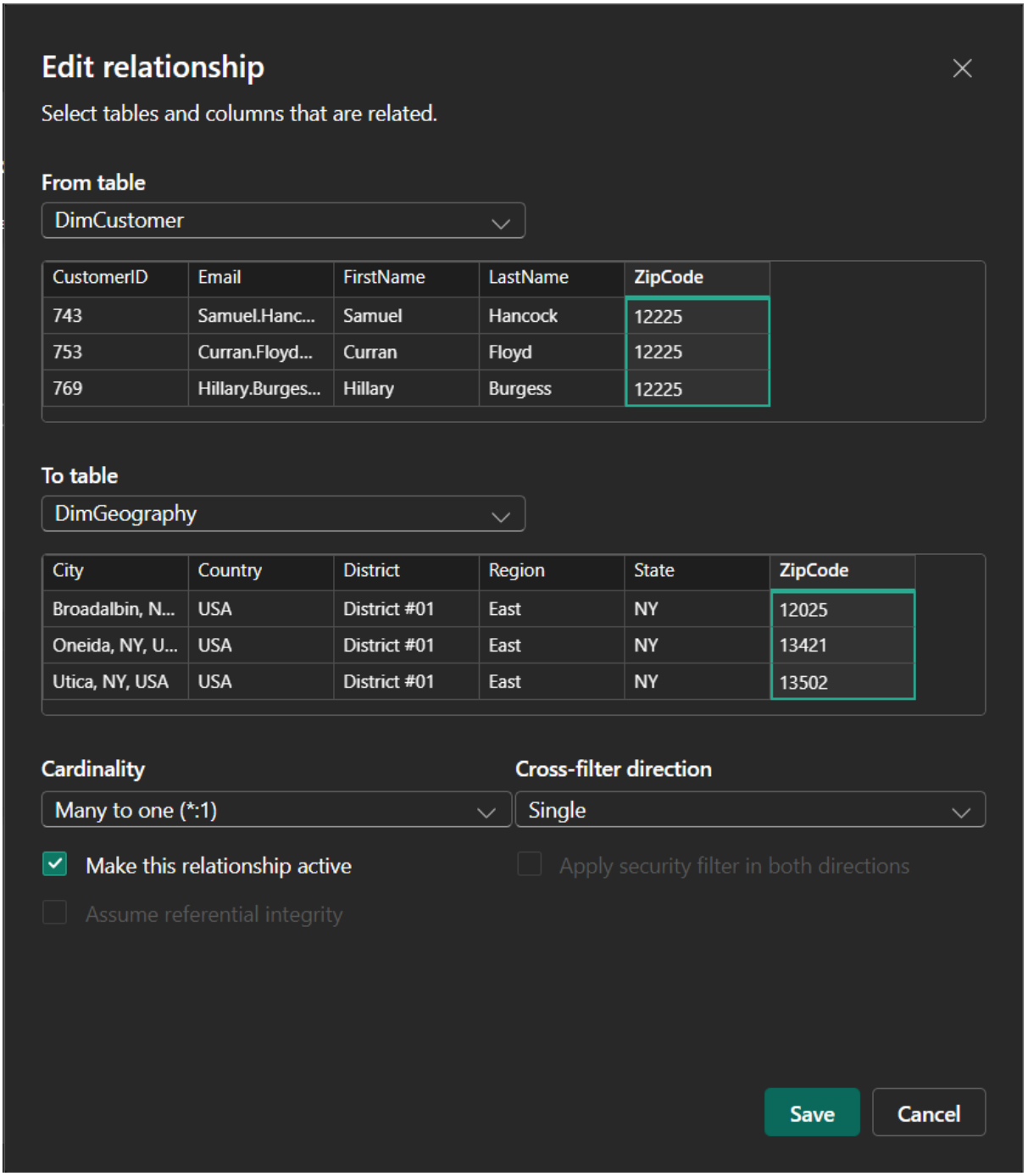


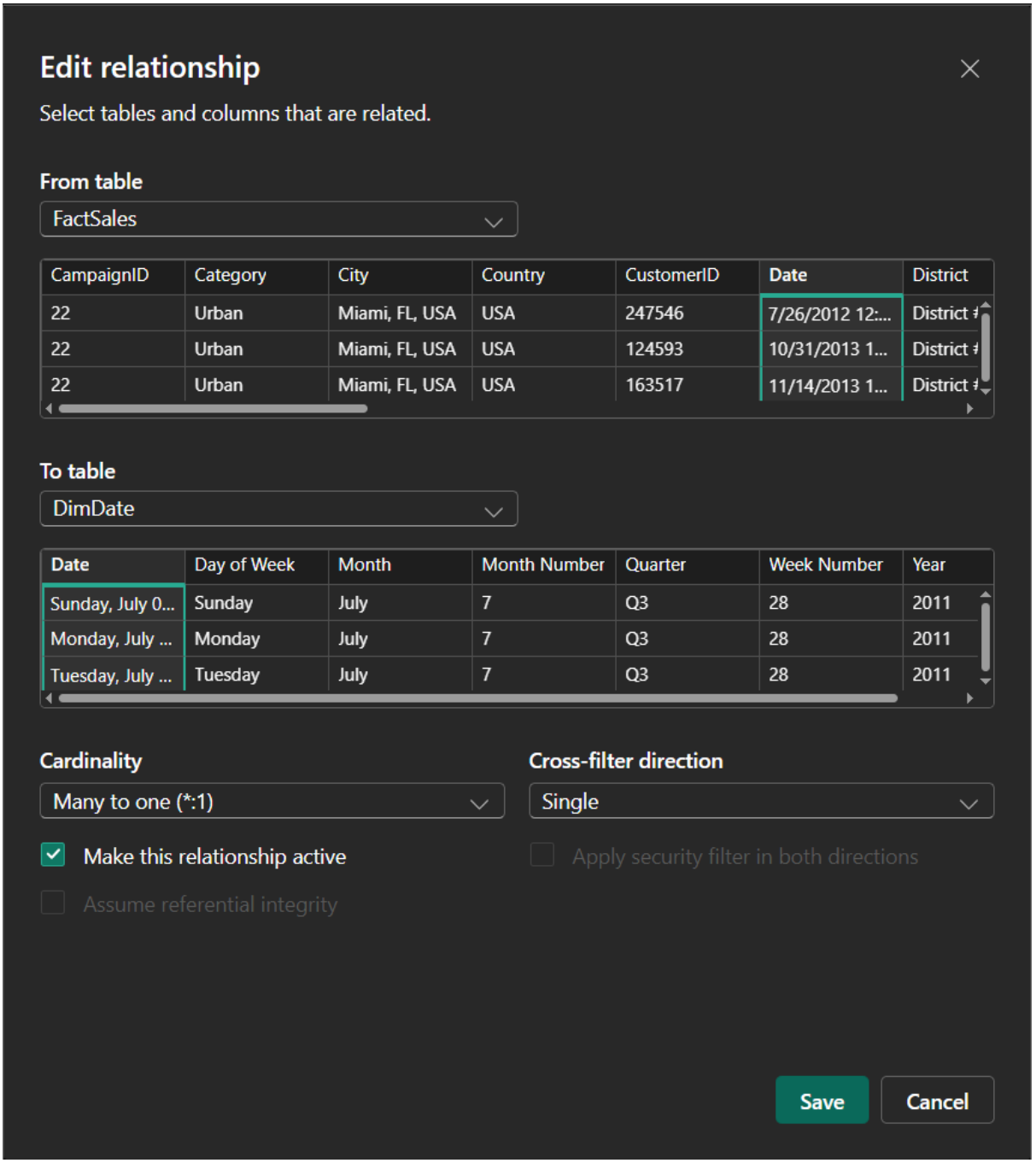
26. Apabila DimGeography memiliki Relationship langsung dengan FactSales, kita delete karena DimGeography merupakan sub dimensional dari DimCustomer dengan cara klik kanan jalurnya > Delete

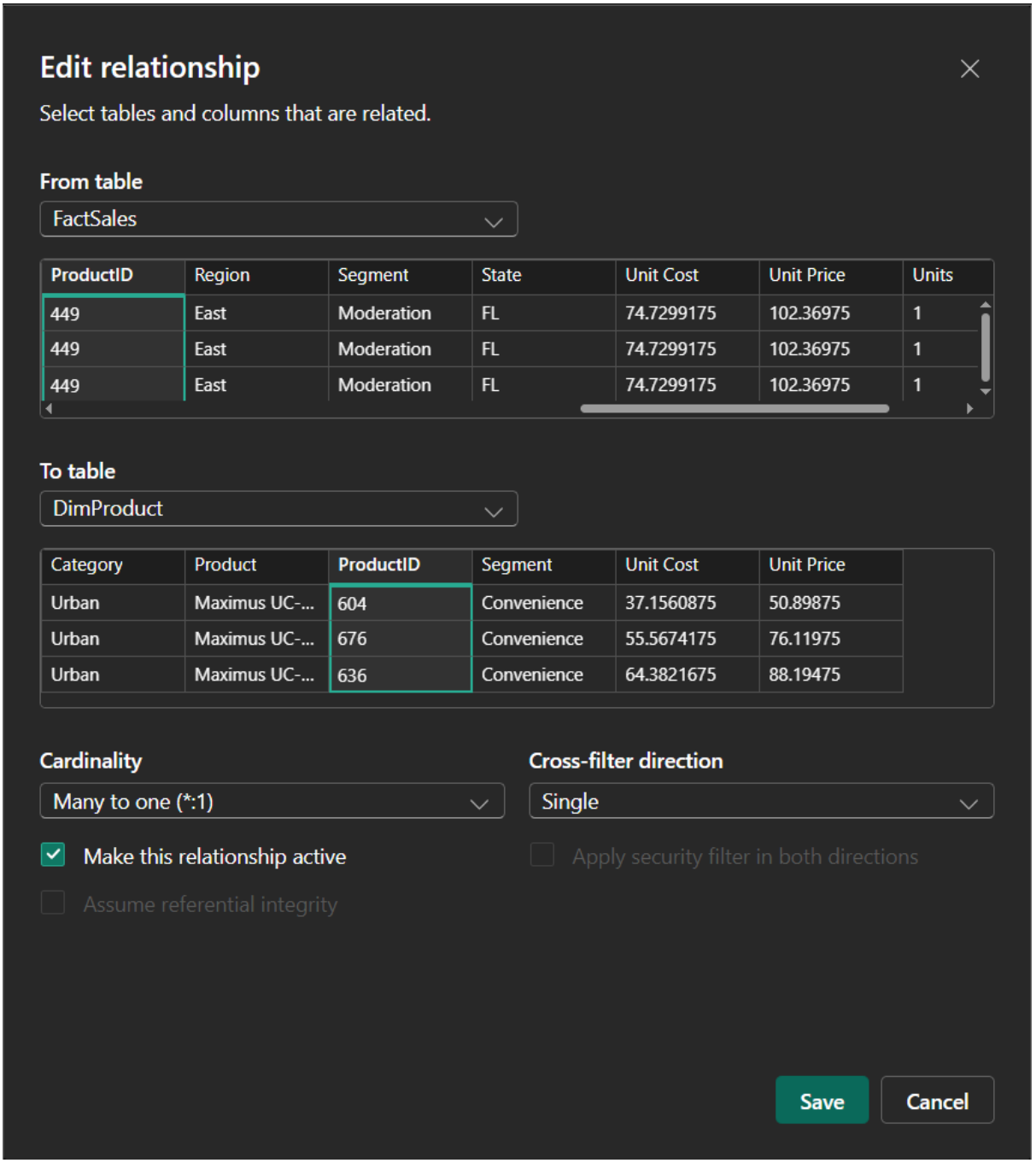


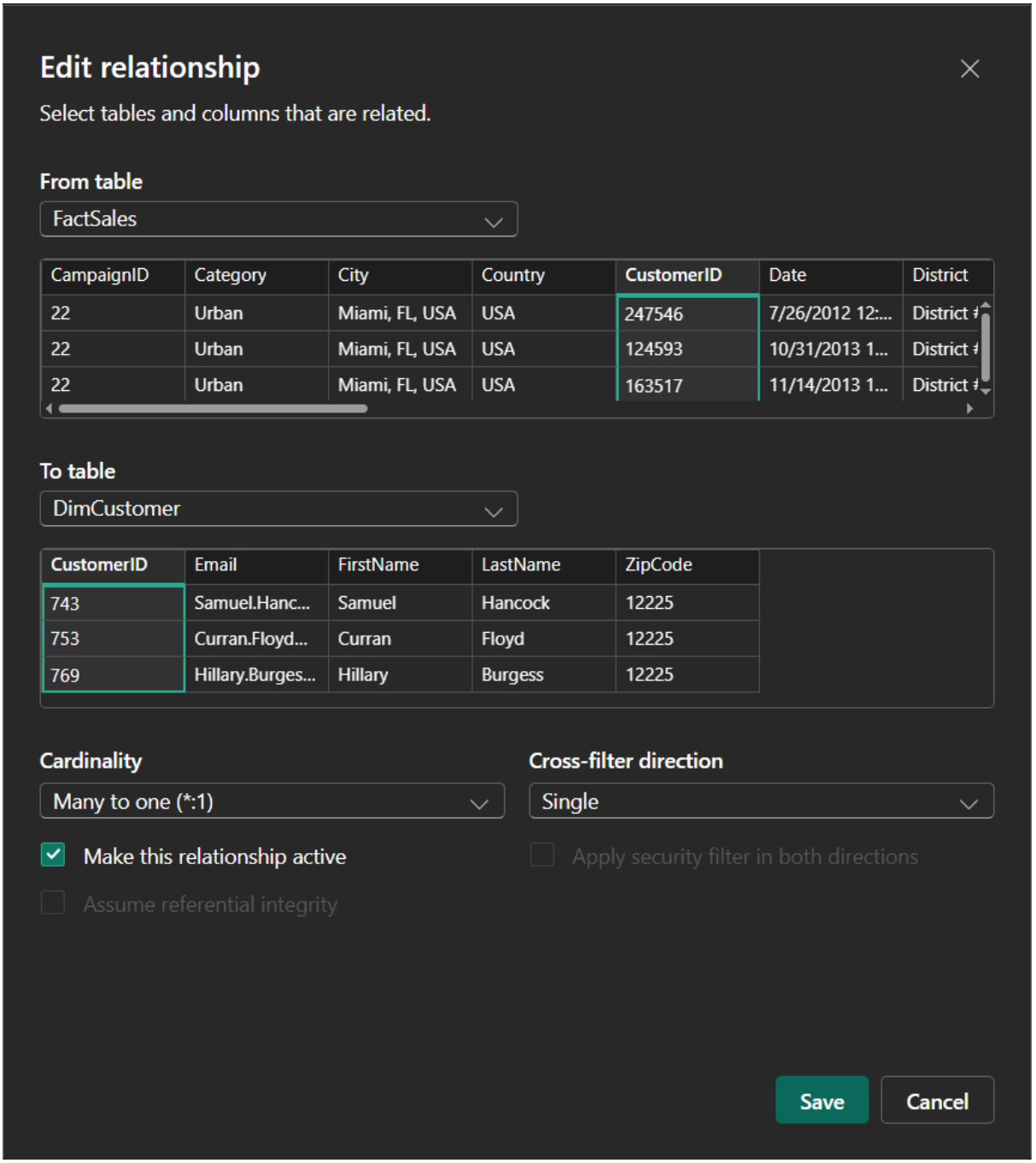
27. Lalu jalur relationship DimGeography dengan DimCustomer diklik kanan > Properties, muncul Edit relationship kemudian kita cukup klik centang Make this relationship active lalu klik Ok.


Untitled - Power BI Desktop
Search
Sign in
File
Home
Help
Share
Paste
Cut
Copy
Clipboard
Get data
Excel workbook
OneLake catalog
SQL Server
Enter data
Dataverse
Recent sources
Data
Transform data
Refresh
Queries
Manage relationships
Relationships
New measure
New column
New table
Calculation group
Calculations
Manage roles
View as
Security
Q&A setup
Language
Linguistic schema
Q&A
Sensitivity
Publish
Share
FactSales
CampaignID
Category
City
Country
CustomerID
Date
District
Email Name
Manufacturer
Collapse
DimGeography
City
Country
District
Region
State
ZipCode
Collapse
DimCustomer
CustomerID
Email
Delete
Properties
Collapse
DimDate
Date
Day of Week
Month
Month Number
Quarter
Week Number
Year
Collapse
DimProduct
Category
Product
ProductID
Segment
Unit Cost
Unit Price
Collapse
Properties
Data
All tables
65%

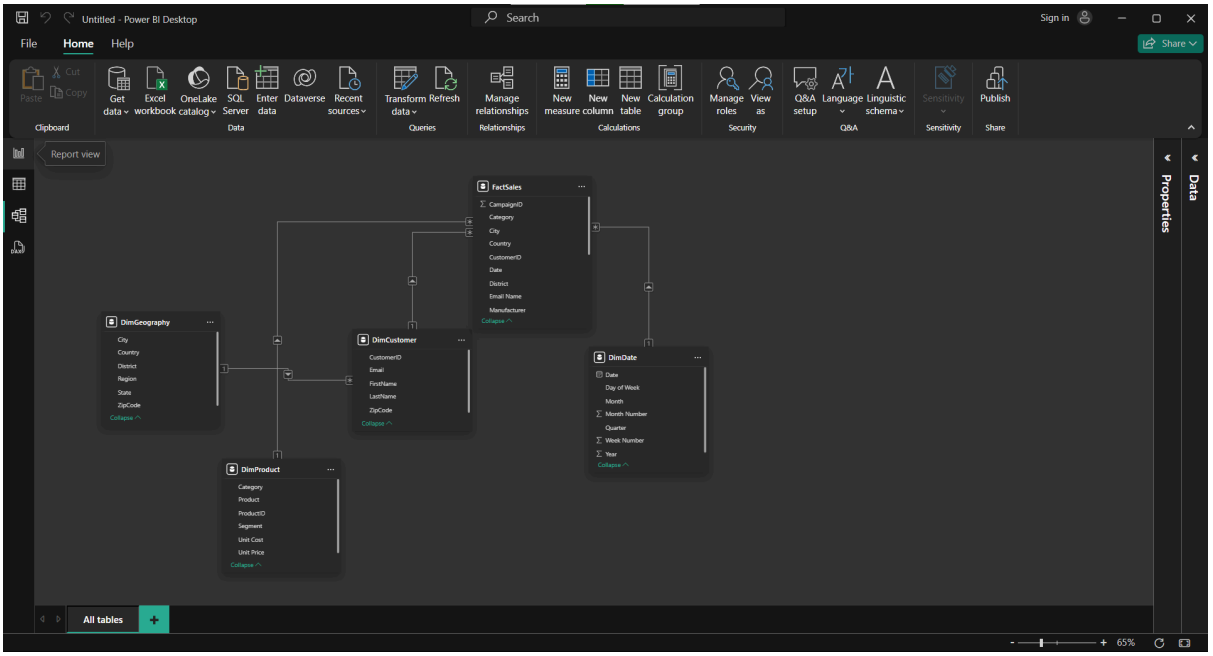

Edit relationship
Select tables and columns that are related.
From table
DimCustomer
CustomerID | Email | FirstName | LastName | ZipCode
743 | Samuel.Hanc... | Samuel | Hancock | 12225
753 | Curran.Floyd... | Curran | Floyd | 12225
769 | Hillary.Burges... | Hillary | Burgess | 12225
To table
DimGeography
City | Country | District | Region | State | ZipCode
Broadalbin, N... | USA | District #01 | East | NY | 12025
Oneida, NY, U... | USA | District #01 | East | NY | 13421
Utica, NY, USA | USA | District #01 | East | NY | 13502
Cardinality
Many to one (*:1)
Cross-filter direction
Single
Make this relationship active
Apply security filter in both directions
Assume referential integrity
Save
Cancel


Edit relationship
Select tables and columns that are related.
From table
FactSales
CampaignID | Category | City | Country | CustomerID | Date | District
22 | Urban | Miami, FL, USA | USA | 247546 | 7/26/2012 12:... | District
22 | Urban | Miami, FL, USA | USA | 124593 | 10/31/2013 1... | District
22 | Urban | Miami, FL, USA | USA | 163517 | 11/14/2013 1... | District
To table
DimDate
Date | Day of Week | Month | Month Number | Quarter | Week Number | Year
Sunday, July 0... | Sunday | July | 7 | Q3 | 28 | 2011
Monday, July ... | Monday | July | 7 | Q3 | 28 | 2011
Tuesday, July ... | Tuesday | July | 7 | Q3 | 28 | 2011
Cardinality
Many to one (*:1)
Cross-filter direction
Single
Make this relationship active
Apply security filter in both directions
Assume referential integrity
Save
Cancel

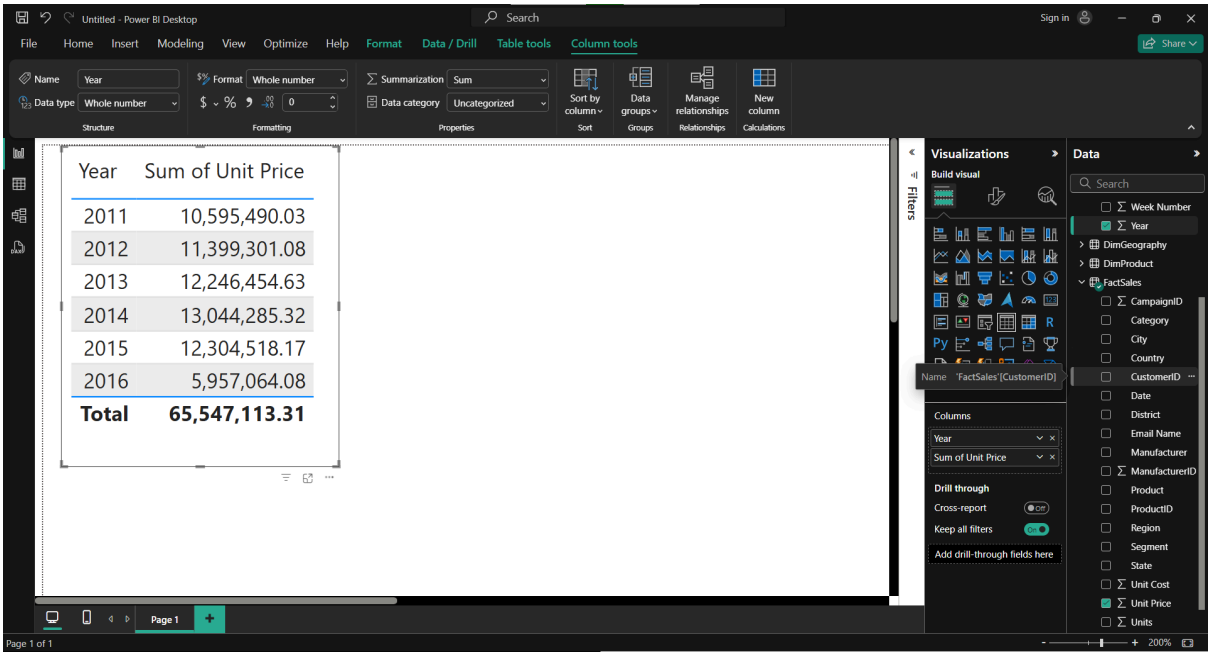

Edit relationship
Select tables and columns that are related.
From table
FactSales
ProductID | Region | Segment | State | Unit Cost | Unit Price | Units
449 | East | Moderation | FL | 74.7299175 | 102.36975 | 1
449 | East | Moderation | FL | 74.7299175 | 102.36975 | 1
449 | East | Moderation | FL | 74.7299175 | 102.36975 | 1
To table
DimProduct
Category | Product | ProductID | Segment | Unit Cost | Unit Price
Urban | Maximus UC-... | 604 | Convenience | 37.1560875 | 50.89875
Urban | Maximus UC-... | 676 | Convenience | 55.5674175 | 76.11975
Urban | Maximus UC-... | 636 | Convenience | 64.3821675 | 88.19475
Cardinality
Many to one (*:1)
Cross-filter direction
Single
Make this relationship active
Apply security filter in both directions
Assume referential integrity
Save
Cancel



28. Setelah relasi sudah diperbaiki, kita beralih ke Report View

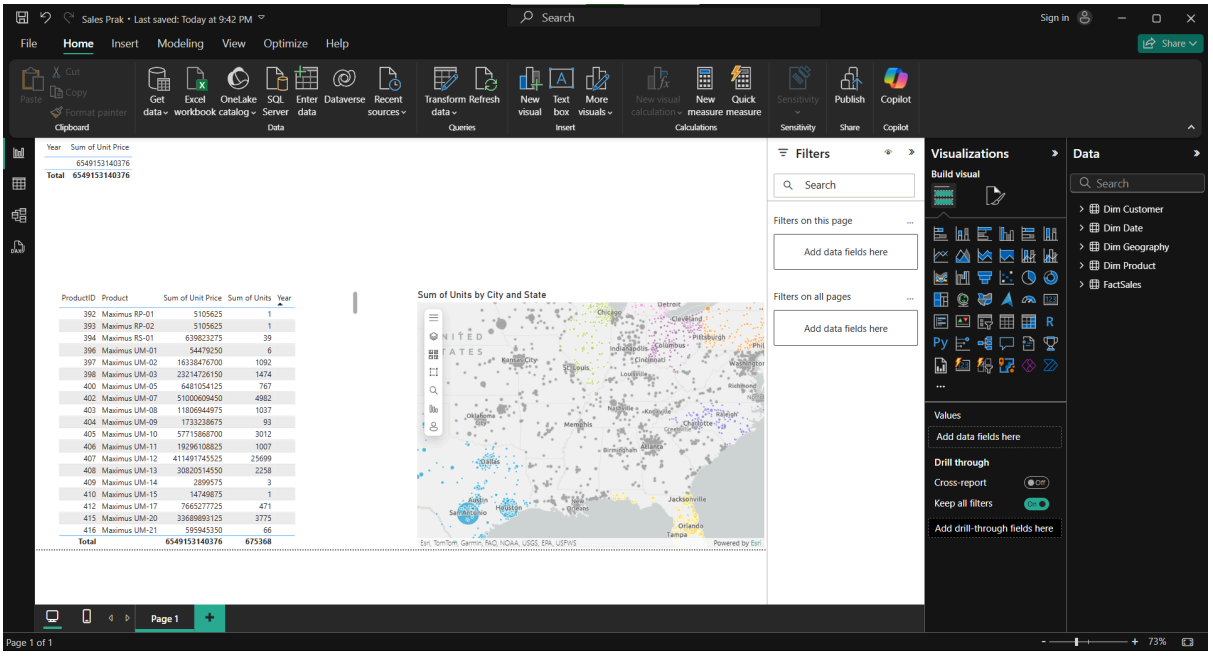


29. Kita buat visualisasi sederhana dengan cara klik visual table lalu pilih kolom Year di DimDate dan Unit Price di FactSales. Atur kolom Year dengan pilih don't summarize lalu hasil seperti screenshoot dibawah



TUGAS

Buatkan visulasisasi seperti Screenshoot dibawah ini sebagai bukti bahwa kita sudah melakukan proses ETL dan modeling data (map menggunakan ArcGIS Maps for Power BI)



1. **Soal Latihan**

   Soal:

   1. Apa perbedaan OLAP dengan OLTP?
   2. Apa yang dimaksud dengan ETL?

   Jawaban:

   1. OLAP (Online Analytical Processing) adalah sistem yang digunakan untuk analisis data dan pengambilan keputusan. OLAP berfokus pada query kompleks dan agregasi data, serta biasanya menggunakan skema multidimensi seperti star atau snowflake schema. Contoh penggunaannya seperti laporan penjualan tahunan atau analisis tren. OLTP (Online Transaction Processing) adalah sistem yang digunakan untuk pengolahan transaksi secara real-time. Fokus utamanya adalah pada efisiensi transaksi seperti insert, update, dan delete. Contoh penggunaannya adalah sistem kasir, sistem pemesanan tiket, atau aplikasi perbankan.
   2. ETL merupakan singkatan dari Extract, Transform, Load, yaitu proses penting dalam pengolahan data yang digunakan untuk memindahkan data dari berbagai sumber ke dalam sistem penyimpanan terpusat seperti data warehouse. Proses ini dimulai dengan extract (mengekstraksi) data dari sumber yang berbeda, baik itu database, file, atau layanan web. Setelah itu, data tersebut masuk ke tahap transform (transformasi) di mana data dibersihkan, disesuaikan formatnya, dihapus data duplikatnya, dan disusun agar sesuai dengan struktur yang dibutuhkan untuk analisis. Terakhir, data yang sudah siap

dimasukkan atau load ke dalam sistem tujuan seperti data warehouse atau dashboard analitik. ETL menjadi fondasi utama untuk mendukung pengambilan keputusan berbasis data, terutama dalam konteks business intelligence dan analisis data.

2. Kesimpulan

a. Dalam pengerjaan praktikum Data Warehouse, kita harus benar-benar teliti dalam menginputkan suatu fungsi untuk menampilkan suatu keluaran pada layar dengan sesuai.

b. Kita dapat mengetahui bahwa proses ETL dan data modeling menggunakan Power BI sangat penting dalam membentuk struktur data yang efisien dan mudah dianalisis. Praktikum ini membantu memahami bagaimana data mentah dari sistem OLTP dapat diubah menjadi bentuk OLAP yang siap digunakan dalam visualisasi. Pembuatan tabel fact dan dimension, serta pengaturan relasi antar tabel, sangat krusial untuk mendukung akurasi dalam analisis. Selain itu, penggunaan tools seperti Power Query dan fitur model di Power BI memberikan pemahaman praktis tentang alur data, mulai dari ekstraksi hingga visualisasi akhir.

3. Cek List (✔)

No	Elemen Kompetensi	Penyelesaian	
		Selesai	Tidak Selesai
1.	Elkom 1	✔	
2.	Tugas	✔	

4. Formulir Umpan Balik

No	Elemen Kompetensi	Waktu Pengerjaan	Kriteria
1.	Elkom 1	10 Menit	menarik
2.	Tugas	10 Menit	menarik

Keterangan:
1. Menarik
2. Baik

3. Cukup
4. Kurang